Stopped Experiments: From Bayes-Risk Contraction to Minimax Regret

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Abstract

We study the regret required to reduce posterior Bayes risk by a fixed factor. A stopped experiment runs until this contraction occurs or the experiment is abandoned; its cost is the risk scale times the expected regret incurred, divided by the probability of contraction. Uniformly selectable upper bounds on this cost compose into worst-prior Bayes-regret bounds, whereas pointwise lower bounds yield calibrated finite-horizon lower bounds. For finite latent models with finite action sets, and for compact Gaussian-feedback loss models, measurable selection and Bayes–minimax closure give the exponent identity

$$\alpha^{\mathrm{M}} = \alpha^{\mathrm{B}} = \frac{1}{2}\alpha^{\mathrm{stop}}.$$

For top-k aggregate-feedback bandits with $2k$ independent Gaussian arms, this method yields minimax regret $\widetilde{\Theta}(\min\{kT, k^{3/2}\sqrt{T}\})$, closing the dense independent-Gaussian slice of a known $\sqrt{k}$ gap. For bandit convex optimization, the full-class regret-exponent gap is equivalent to a stopped-cost gap; on a dimple family, an observation-adaptive experiment has cost $\widetilde{O}(d^2)$, while every one-query or fresh-direction experiment costs $\Omega(d^3)$. We also calculate the stopped cost for a deterministic preference-feedback chain under probit and Bradley–Terry comparisons, obtaining matching fixed-scale value-regret bounds up to logarithms.

1 Introduction

How much regret must be spent before observations make a decision problem meaningfully easier? A fixed-confidence guarantee may count the observations needed to identify a good action, but it need not control the regret incurred by those queries or a policy that learns only on favorable transcript branches. Nor is a single query always the right unit: an early observation may matter chiefly because it determines which later queries should be made. We therefore study complete adaptive experiments whose stopping target is a reduction in posterior decision risk.

Stopped risk-contraction experiments. We begin with a fixed latent environment ω. An action a incurs a nonnegative decision gap $\Delta_\omega(a)$ and produces an observation. If Q_t is the posterior after t observations, its Bayes decision risk is

$$r(Q_t) = \inf_a \mathbb{E}_{Q_t}\Delta_\omega(a).$$

At a state where this risk is of order b, an adaptive experiment runs until it is abandoned or first reaches $r(Q_t) \le \gamma b$. If N is its duration and M the first-hit event, we define the pointwise stopped cost by

$$\mathsf{SC}_b(Q) = \inf_{\text{stopped experiments}} \frac{b\,\mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})}. \tag{1}$$

We call this a stopped risk-contraction experiment, or simply a stopped experiment. Thus $\mathsf{SC}_b(Q)$ is the regret paid per unit probability of a decision-relevant reduction in risk, after scaling by the current risk level. The definition introduces no auxiliary information budget or dimension-dependent normalization. A separate query charge is unnecessary: every query before the hit has posterior expected gap at least γb, so $\mathbb{E}R_N \geq \gamma b\,\mathbb{E}N$.

Two-sided regret transfer. The paper's central result transfers between stopped risk contraction at one posterior and scale and finite-horizon regret. For an upper bound, one constructs at each risk scale an adaptive experiment that contracts posterior risk at controlled regret cost; the theorem concatenates these designs across scales. For a lower bound, one strengthens a testing or estimation inequality so that it applies to stopped transcripts and remains linear in the probability of contraction; the theorem then balances the branch that contracts with the branch that continues to incur risk. The transfer is not automatic. Ordinary sample complexity neither prices the decisions used to collect data nor controls policies that learn only on favorable transcript branches.

Let $\mathsf{SC}^{\mathrm{pt}}$ optimize separately at each posterior state and let $\mathsf{SC}^{\mathrm{unif}}$ require one measurable state-dependent experiment rule. We prove

$$\mathfrak{R}^{\mathrm{B}}(d,T) \lesssim_\gamma 1 + \sqrt{\mathsf{SC}^{\mathrm{unif}}(d,T)T},$$

and, whenever $c < \mathsf{SC}_b(Q)$, a lower bound of order $\sqrt{cn}$ at $n \asymp_\gamma c/b^2$. The upper proof joins experiments over geometric risk scales. The lower proof stops an arbitrary regret policy at its first contraction and balances the hitting and non-hitting branches. When statewise experiments can be selected measurably and worst-prior Bayes regret equals minimax regret, the two directions close:

$$\alpha^{\mathrm{M}} = \alpha^{\mathrm{B}} = \frac{1}{2}\alpha^{\mathrm{stop}}.$$

We verify these closure conditions for finite latent-model problems with finite action sets and for Gaussian-feedback loss classes on compact action spaces. The equalities concern polynomial dimension exponents, not a same-horizon identity.

The results below play different roles. The exponent identity is the general characterization. The dense top-k theorem is the strongest new finite-horizon minimax consequence, and the dimple construction is the sharpest structural separation. The preference model then shows how the same method passes through a nonlinear binary channel while charging both policies used to obtain a comparison.

1. For top-k full-bandit feedback with $n = 2k$ independent Gaussian arms, we calculate $\mathsf{SC}_\varepsilon(Q_0) = \Theta_\gamma(k)$ on a paired family and obtain

$$\mathfrak{R}^\star_{2k,k}(T) = \widetilde{\Theta}\Big(\min\{kT, k^{3/2}\sqrt{T}\}\Big)\,.$$

   The upper bound is the Hadamard-design guarantee of Rejwan and Mansour (2020). This closes, for the dense independent-Gaussian model, the factor-$\sqrt{k}$ gap displayed in their Corollary 8. The same dense rate was previously known for grouped action sets, so Table 1 identifies precisely what changes here and what remains open.

2. For a deterministic H-stage chain with k binary decisions, a probit or Bradley–Terry trajectory comparator, and $\alpha = \beta\varepsilon/H$, we prove

$$\mathsf{SC}^{\mathrm{pref}}_b(Q_0) = \Theta_{\gamma,F}\big(b^2\{k + \alpha^{-2}\}\big)\,, \qquad b = \varepsilon k/H.$$

This gives matching fixed-scale pair-average value regret up to a logarithm. For comparisons of noisy aggregate trajectory scores with $H = k$, it specializes to $\mathsf{SC}_\varepsilon^{\mathrm{pref}} = \Theta_\gamma(k)$ and the same dense $k^{3/2}\sqrt{T}$ envelope as aggregate feedback. Every label uses two episodes, and both episode losses are charged.

3. For the full BCO class, the closure theorem restates the current gap as $5/2 \le \alpha_{\mathrm{stop}} \le 3$. On the dimple family of Bakhtiari et al. (2025), at the spherical-prior initial state we further prove

$$\mathsf{SC}_b^{[1]},\ \mathsf{SC}_b^{\mathrm{fresh}} = \Omega(d^3), \qquad \mathsf{SC}_b = \widetilde{O}(d^2)$$

at $\varepsilon = d^{-1}$. This separates experiment classes; it is not an unrestricted lower bound for that family.

The results require nonnegative additive gaps, a posterior that is sufficient for future decisions, and repetition of the same latent decision problem after each observation. Convexity and Gaussian observations are needed only in the measurable-selection and BCO sections.

1.1 Related work

Regret complexity and sequential design. Information ratios, AIR, and DEC turn one-step regret–information or decision–estimation comparisons into sequential regret bounds (Russo and Van Roy, 2016, 2018; Lattimore and György, 2021; Foster et al., 2021, 2022, 2023; Xu and Zeevi, 2025; Neu et al., 2024); chained ratios organize such comparisons across action scales (Gouverneur et al., 2024). These methods may re-optimize after every observation. The difference is therefore not whether the policy is adaptive, but the primitive certificate: stopped cost evaluates an entire adaptive experiment, including how early observations determine later queries and stopping, rather than assigning a value to one query at a time. Interactive Le Cam, Fano, and Assouad inequalities provide lower-bound tools for the resulting adaptive transcript laws (Chen et al., 2024). The construction is also related to sequential experimental design. Blackwell comparison and Bayesian theories of experiments value observations through downstream decisions (Blackwell, 1953; Lindley, 1956; DeGroot, 1962), while sequential decision theory and active testing choose experiments and stopping rules (Arrow et al., 1949; Chernoff, 1959; Naghshvar and Javidi, 2013). Those theories motivate the experiment language, but their standard sample or terminal-decision costs are not used as black-box inputs here. The required certificate is more specific: it charges the decision loss of every query, stops at a fixed-factor reduction in posterior Bayes risk, and retains the probability of reaching that set.

Risk contraction and adaptive experiments. Several approaches avoid exact identification by seeking only enough information to choose an action whose loss is within a prescribed tolerance of optimal (Dong and Van Roy, 2018; Russo and Van Roy, 2022; Arumugam and Van Roy, 2021; Bubeck and Sellke, 2023). This is the relevant meaning of a near-optimal target: at tolerance b, one need not identify the environment or a unique optimizer. Here the target is posterior—success means $r(Q_N) \le \gamma b$—and different transcripts may end with different near-optimal actions. Related pointwise Gaussian analysis localizes complexity around an individual target rather than the full ambient class (Xu, 2026b). In this paper, "pointwise" has the narrower technical meaning that the experiment is optimized separately at each posterior state; measurable selection asks whether these statewise choices can be implemented by one rule. Finally, Maiti et al. (2026) show in linear best-arm identification that early observations can reveal which later probes are useful. The present paper uses neither result as a technical premise: they clarify, respectively, why the relevant target is decision-level risk rather than exact model recovery and why a complete adaptive experiment can be strictly cheaper than one-query certificates.

Closest application precedents. The Bellman-sufficient framework of Xu (2026a) accommodates physical states, contexts, and other history-dependent variables. The present theory treats the repeated latent-decision specialization in which the posterior is sufficient and the same nonnegative gap recurs; these restrictions make posterior risk persistent enough for the two-sided transfer. On posterior space, the stopped experiment is a stochastic shortest-path problem (Bertsekas and Tsitsiklis, 1991). Application-specific comparisons accompany the corresponding results. Table 1 shows that the top-k theorem closes the dense $n = 2k$, independent-Gaussian slice of the gap displayed by Rejwan and Mansour (2020), while leaving general n open. The preference calculation extends the same paired statistical core through a binary comparison channel; it is not a general tabular-RL rate (Xu et al., 2020; Saha et al., 2023; Wang et al., 2023). For BCO, the main model-specific conclusion is the polynomial separation, on the dimple family of Bakhtiari et al. (2025), between observation-adaptive experiments and the one-query or fresh-direction classes.

2 Stopped experiments and regret transfer

2.1 The repeated latent decision problem

We first study a repeated decision problem with a fixed latent environment; convexity is not required. This is the static-state specialization of the Bellman-sufficient framework of Xu (2026a): the current posterior is the pre-action state, the same action problem recurs after every observation, and the action gap is a nonnegative one-step Bellman loss. General stateful control may also possess a Bellman-sufficient state, but need not satisfy the persistence property used below to concatenate experiments across risk scales.

Let Ω, $\mathcal{A}$, and $\mathcal{O}$ be standard Borel environment, action, and observation spaces, and let $\Delta_\omega(a) \in [0, 1]$ be a nonnegative action gap. Querying a produces an observation in $\mathcal{O}$ from a kernel $P_\omega(\cdot \mid a)$. The environment $\omega \sim Q$ is drawn once and remains fixed. A policy chooses A_t from the pre-action history H_{t-1}, observes O_t, and incurs $\Delta_\omega(A_t)$. We include all independent exogenous policy randomization in H_0 and write $\mathcal{H}_t = \sigma(H_t)$ for the completed history filtration.

For a posterior Q, define the Bayes risk

$$r(Q) = \inf_{a\in\mathcal{A}} \mathbb{E}_Q \Delta_\omega(a).$$

We assume the posterior is a sufficient controlled state, r is measurable, and measurable ϵ-Bayes actions exist. These are standard in compact Borel models and are verified below for Gaussian-feedback loss classes on compact action spaces. The theorem applies to static latent-model decision problems of the DMSO type (Foster et al., 2021). Neither the action space nor the map $a \mapsto \Delta_\omega(a)$ is assumed to be convex. Nonnegativity is essential. If one-step increments can be signed, stopping may discard later negative increments that would offset earlier positive ones, and regret need not control the number of elapsed queries. Such problems require a separate query charge or a lower-bounded Bellman loss.

The repeated-decision structure implies that posterior Bayes risk is a nonnegative supermartingale. Indeed,

$$\mathbb{E}[r(Q_{t+1}) \mid \mathcal{H}_t] \le \inf_{a\in\mathcal{A}} \mathbb{E}[\mathbb{E}\{\Delta_\omega(a) \mid \mathcal{H}_{t+1}\} \mid \mathcal{H}_t] = r(Q_t). \tag{2}$$

This fact, rather than convexity or a particular observation law, controls returns to a risk scale in the upper bound.

Fix

$$0 < \gamma < \vartheta < 1. \tag{3}$$

A pair $s = (Q, b)$ is *active* when

$$\vartheta b < r(Q) \le b.$$

Starting from an active state, let

$$\tau_b = \inf\{t \ge 1 : r(Q_t) \le \gamma b\} \tag{4}$$

be the first posterior-risk contraction.

Definition 2.1 (Stopped cost)**.** *An admissible experiment consists of an adaptive query policy and a positive, possibly infinite, $(\mathcal{H}_t)$-stopping time σ. It runs for*

$$N = \sigma \wedge \tau_b$$

rounds and succeeds on $\mathsf{M} = \{\tau_b \le \sigma\}$. We require $N < \infty$ almost surely. Expectations below are initially extended-valued; Lemma 2.2 shows that every experiment with finite displayed ratio has $\mathbb{E}N < \infty$. Define

$$\mathsf{SC}_b(Q) := \inf_{\pi,\sigma} \frac{b\,\mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})}, \qquad R_N = \sum_{t=1}^{N} \Delta_\omega(A_t), \tag{5}$$

with ratio $+\infty$ when the success probability is zero.

The first-hit convention loses nothing: after reaching the lower-risk set, additional observations cannot improve the success event and only add nonnegative regret. It also ensures that every charged query is made while risk exceeds γb, which is what turns elapsed queries into regret. Voluntary stopping is essential because an adaptive design may abandon an unpromising branch. Dividing by $\mathbb{P}_Q(\mathsf{M})$ prevents rare successful branches from making such abandonment artificially free.

Lemma 2.2 (Query cost is already charged)**.** *Every admissible experiment satisfies*

$$\mathbb{E}_Q R_N \ge \gamma b\, \mathbb{E}_Q N. \tag{6}$$

Consequently,

$$\mathsf{SC}_b(Q) \le \inf_{\pi,\sigma} \frac{b^2 \mathbb{E}_Q N + b\mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})} \le (1 + \gamma^{-1})\mathsf{SC}_b(Q).$$

The proof is a stopped-sum application of the defining inequality $r(Q_{t-1}) > \gamma b$ before the first hit; see Appendix A.

2.2 Bellman representation of the stopped ratio

The stopped cost is a stochastic shortest-path problem on posterior space. For a query cap L and price $q > 0$, put $W_0^q = 0$ and

$$\begin{aligned} J_k^q(Q, b) = \inf_{a \in \mathcal{A}} \Big\{ & b\,\mathbb{E}_Q \Delta_\omega(a) \\ & + \int \left[-q\,\mathbf{1}\{r(Q^{a,o}) \le \gamma b\} + \mathbf{1}\{r(Q^{a,o}) > \gamma b\} W_{k-1}^q(Q^{a,o}, b) \right] P_Q(do \mid a) \Big\}, \\ W_k^q(Q, b) = \min\{0, & J_k^q(Q, b)\}. \end{aligned} \tag{7}$$

The root uses J^q_L, so the experiment makes at least one query; W^q_k permits voluntary stopping on a non-hitting branch. If $\mathsf{SC}_{b,L}$ is the infimum in (5) over experiments with $N \le L$ almost surely, then

$$\mathsf{SC}_{b,L}(Q) < q \quad \iff \quad J^q_L(Q,b) < 0, \qquad \mathsf{SC}_b(Q) = \inf_{L\ge 1} \mathsf{SC}_{b,L}(Q). \tag{8}$$

The first equivalence follows by minimizing $b\,\mathbb{E}R_N - q\,\mathbb{P}(\mathsf{M})$; the second follows by truncating the stopping time. The continuation value at $Q^{a,o}$ credits an early observation for improving later queries, even when that observation does not itself reach the contraction set.

2.3 Pointwise and uniformly selectable stopped-cost profiles

Let $\mathfrak{D} = (\mathfrak{D}_d)$ be a sequence of such decision problems. For a scale window $B^{-1} \le b \le 1$, write $\mathsf{S}_{\mathfrak{D}}(d,B)$ for the active posterior states. Define

$$\mathsf{SC}^{\mathrm{pt}}_{\mathfrak{D}}(d,B) = \sup_{(Q,b)\in \mathsf{S}_{\mathfrak{D}}(d,B)} \inf_{\mathcal{E}} \frac{b\,\mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})}, \tag{9}$$

$$\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d,B) = \inf_{\Sigma \text{ universally measurable}} \sup_{(Q,b)\in \mathsf{S}_{\mathfrak{D}}(d,B)} \frac{b\,\mathbb{E}^{\Sigma(Q,b)}_Q R_N}{\mathbb{P}^{\Sigma(Q,b)}_Q(\mathsf{M})}. \tag{10}$$

The supremum of an empty set is zero. Always $\mathsf{SC}^{\mathrm{pt}} \le \mathsf{SC}^{\mathrm{unif}}$. The first quantity chooses the best experiment separately at each state. The second requires all statewise choices to form one universally measurable experiment rule.

Let

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d,T) = \sup_Q \inf_\pi \mathbb{E}^\pi_Q R_T$$

denote worst-prior Bayes regret. If $\mathfrak{D}_d$ contains several known action spaces, the definition also takes the supremum over those spaces.

Theorem 2.3 (Local-to-global regret transfer)**.** *For every $T \ge 2$ and $T^{-1} \le \eta < 1/2$,*

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d,T) \le \min\left\{T,\ C_{\gamma,\vartheta}\frac{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d,T)}{\eta} + 2T\eta\right\}. \tag{11}$$

In particular,

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d,T) \le C_{\gamma,\vartheta} \min\left\{T,\ 1 + \sqrt{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d,T)\,T}\right\}. \tag{12}$$

Conversely, for every active state (Q,b), every $0 < c < \mathsf{SC}_b(Q)$, and every $n \ge 1$,

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d,n) \ge \inf_\pi \mathbb{E}^\pi_Q R_n \ge \frac{\gamma c b n}{c + \gamma b^2 n} \ge \frac{1}{2}\min\left\{\gamma b n, \frac{c}{b}\right\}. \tag{13}$$

In particular, if $0 < c < \mathsf{SC}^{\mathrm{pt}}_{\mathfrak{D}}(d,B)$, some active state with $B^{-1} \le b \le 1$ satisfies this premise. If $c/(\gamma b^2) \ge 2$, then at

$$n = \left\lfloor \frac{c}{\gamma b^2} \right\rfloor$$

one has

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d,n) \ge \frac{\sqrt{\gamma}}{4}\sqrt{cn}. \tag{14}$$

How the multiscale transfer works. For the upper bound, invoke a stopped experiment at the active geometric risk scale and play a Bayes action below scale 2η. After a hit at scale b_j, the risk supermartingale and Doob's inequality bound the probability of returning above ϑb_j by γ/ϑ. Thus the expected number of visits to each scale is geometric, and summing the experiment costs gives $O_{\gamma,\vartheta}(\mathsf{SC}^{\text{unif}}/\eta)$; low-risk rounds cost at most $2T\eta$.

The lower direction contains the essential conversion. Stop an arbitrary n-round policy at $N = \tau_b \wedge n$ and write $p = \mathbb{P}_Q(\tau_b \leq n)$. If $\mathsf{SC}_b(Q) > c$, the same policy must satisfy

$$\mathbb{E}_Q R_n \geq \frac{c}{b}\, p, \qquad \mathbb{E}_Q R_n \geq \gamma b n(1-p). \tag{15}$$

The first inequality prices the branch that reaches the contraction set; the second prices the branch that remains at risk scale b. Minimizing their maximum over p gives (13). This is why a constant-confidence sample bound alone is insufficient: it controls neither decision cost nor a policy that contracts only on selected transcripts. Appendix A gives the full proof, including horizon truncation and the geometric-scale concatenation.

2.4 Exponent sandwich and two-sided closure

The finite-sample upper and lower bounds are stated at different risk scales and, on the lower side, at a calibrated horizon. Polynomial exponents suppress these calibration differences while preserving the distinction between statewise existence and measurable implementation. Define

$$\alpha_{\mathfrak{D}}^{\text{stop,pt}} = \sup_{m\in\mathbb{N}} \limsup_{d\to\infty} \frac{\log\{1+\mathsf{SC}_{\mathfrak{D}}^{\text{pt}}(d, d^m)\}}{\log d}, \tag{16}$$

$$\alpha_{\mathfrak{D}}^{\text{stop,unif}} = \sup_{m\in\mathbb{N}} \limsup_{d\to\infty} \frac{\log\{1+\mathsf{SC}_{\mathfrak{D}}^{\text{unif}}(d, d^m)\}}{\log d}, \tag{17}$$

$$\alpha_{\mathfrak{D}}^{\text{B}} = \sup_{M\in\mathbb{N}} \limsup_{d\to\infty} \sup_{2\leq T\leq d^M} \frac{\log\{1+\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d,T)/\sqrt{T}\}}{\log d}. \tag{18}$$

If a frequentist minimax value $\mathfrak{R}_{\mathfrak{D}}^{\text{M}}$ is defined, let $\alpha_{\mathfrak{D}}^{\text{M}}$ be given by the same display.

Theorem 2.4 (Regret and stopped exponents)**.** *Every sequence of repeated latent decision problems satisfying the assumptions of this section obeys*

$$\frac{\alpha_{\mathfrak{D}}^{\text{stop,pt}}}{2} \leq \alpha_{\mathfrak{D}}^{\text{B}} \leq \frac{\alpha_{\mathfrak{D}}^{\text{stop,unif}}}{2}. \tag{19}$$

Neither inequality requires convexity or a particular observation law.

Suppose, in addition, that for every finite d, B and every $\epsilon > 0$, the statewise experiments have a universally measurable choice whose ratio at every active (Q, b) is at most $\mathsf{SC}_b(Q) + \epsilon$. Then

$$\mathsf{SC}_{\mathfrak{D}}^{\text{pt}}(d,B) = \mathsf{SC}_{\mathfrak{D}}^{\text{unif}}(d,B) =: \mathsf{SC}_{\mathfrak{D}}(d,B),$$

and

$$\alpha_{\mathfrak{D}}^{\text{B}} = \frac{\alpha_{\mathfrak{D}}^{\text{stop}}}{2}. \tag{20}$$

If finite-horizon Bayes–minimax equality also holds, then

$$\alpha_{\mathfrak{D}}^{\text{M}} = \alpha_{\mathfrak{D}}^{\text{B}} = \frac{\alpha_{\mathfrak{D}}^{\text{stop}}}{2}. \tag{21}$$

Corollary 2.5 (Finite latent-model closure)**.** *Consider a sequence of repeated latent decision problems satisfying the standing assumptions. Suppose that for every d, the latent class Ω_d and action set $\mathcal{A}_d$ are finite, although their cardinalities may grow with d. The observation space may be any standard Borel space, and the observation laws may be arbitrary Borel kernels. Then, for every finite B, T,*

$$\mathsf{SC}^{\mathrm{pt}}_{\mathfrak{D}}(d, B) = \mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d, B), \qquad \mathfrak{R}^{\mathrm{M}}_{\mathfrak{D}}(d, T) = \mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d, T), \tag{22}$$

where

$$\mathfrak{R}^{\mathrm{M}}_{\mathfrak{D}}(d, T) := \inf_{\pi} \sup_{\omega \in \Omega_d} \mathbb{E}^{\pi}_{\omega} R_T.$$

Consequently,

$$\alpha^{\mathrm{M}}_{\mathfrak{D}} = \alpha^{\mathrm{B}}_{\mathfrak{D}} = \frac{1}{2}\alpha^{\mathrm{stop}}_{\mathfrak{D}}. \tag{23}$$

The proof applies Theorem 2.3 on polynomial scale windows and calibrates $n \asymp \mathsf{SC}_b(Q)/b^2$ on the lower side. Details are in Appendix A.

This equality does not require a separate information scale or an assumption that every query is informative. It still requires nonnegative gaps, posterior sufficiency, repetition of the same decision problem, and measurable selection.

3 Aggregate-feedback top-k bandits

We now use the conversion to obtain a finite-horizon minimax result. There are n independent arms with unit-variance Gaussian rewards. At round t, the learner selects $S_t \subset [n]$, $|S_t| = k$, and observes only

$$Y_t = \sum_{j \in S_t} X_{t,j}.$$

The reward is the same sum. Thus this is full-bandit, rather than semi-bandit, feedback. Write $\mathfrak{R}^{\star}_{n,k}(T)$ for minimax expected regret over mean vectors in $[0, 1]^n$.

Theorem 3.1 (Dense top-k aggregate-feedback regret)**.** *There are universal constants $c, C > 0$ such that, for $n = 2k$,*

$$c \min\{kT, k^{3/2}\sqrt{T}\} \le \mathfrak{R}^{\star}_{2k,k}(T) \le C \min\{kT, k^{3/2}\sqrt{T \log(eT)}\}. \tag{24}$$

Consequently the minimax rate is $\widetilde{\Theta}(\min\{kT, k^{3/2}\sqrt{T}\})$.

The upper bound is the minimum of the trivial kT bound and the polynomial-time Hadamard-design CSAR guarantee $O(k\sqrt{nT\log(eT)})$ of Rejwan and Mansour (2020). Their Corollary 8 displays a factor-$\sqrt{k}$ gap, up to logarithms, between this bound and the semi-bandit lower bound and leaves tighter k-dependence for further research. Theorem 3.1 closes the dense $n = 2k$, independent-Gaussian slice of that question. It does not settle the general (n, k) phase diagram.

The rate and hard family have close precedents, so the novelty should be stated narrowly. Theorem 3.1 establishes the lower bound for the full family of k-subsets under independent Gaussian arm noise; earlier results either restricted the feasible sets to one arm per group or used correlated feedback. Moreover, on this dense paired family, any unrestricted query can be reproduced by two one-per-pair queries at exactly twice the regret; see Remark 3.2. Thus the stopped-cost calculation is useful chiefly as a reusable local certificate. This symmetric finite-horizon lower bound also admits direct proofs.

Table 1: Closest results for top-k sum feedback. Rates are in unnormalized selected-sum units. "Independent" means conditionally independent given the fixed instance.

Work	feasible actions	within-round law	result relevant here
Audibert et al. (2014)	one arm from each of k groups	independent Bernoulli coordinates	$\Omega(k\sqrt{nT})$; the same dense scale under the restriction of exactly one arm per group
Cohen et al. (2017)	grouped multitask actions	common Gaussian shock	stronger grouped lower bound from correlated coordinates; not the independent-arm model
Ito et al. (2019)	all k-subsets	strongly correlated binary coordinates	$\Omega(\sqrt{nk^3T})$ in the principal regime; the stronger scale uses correlation
Agarwal et al. (2021)	all k-subsets	possibly correlated, nonlinear aggregate reward	$\Omega(k\sqrt{knT})$ in their model; not an independent additive arm lower bound
Rejwan and Mansour (2020)	all k-subsets	independent sub-Gaussian arms	$O(k\sqrt{nT\log(eT)})$; the matching model and upper bound
This work	all k-subsets, $n = 2k$	independent unit-variance Gaussian arms	$\Omega(k^{3/2}\sqrt{T}) = \Omega(k\sqrt{nT})$, matching up to logarithms

What remains open for general n. A direct extension of the paired proof is not available when $n \gg k$. For example, in a k-group, one-good-arm prior, the posterior-optimal unrestricted k-set may choose several arms from one group and omit another; its risk is governed by the k largest posterior marginals globally, not by a sum of k within-group classification risks. Thus the grouped lower bound of Audibert et al. (2014) does not transfer. A route to the full $k\sqrt{nT}$ lower scale would be a localized stopped adaptive-sensing inequality of the form

$$\mathbb{E}N \gtrsim \frac{pn}{\varepsilon^2}$$

for every aggregate-query experiment whose posterior risk contracts with probability p. The linear dependence on p, required because stopped cost divides by p, is exactly the unproved step. We therefore leave the unrestricted general-(n, k) problem open.

Pair the arms and draw $\theta = (\theta_1, \dots, \theta_k)$ uniformly from $\{-1, +1\}^k$. For $i \in [k]$, set

$$\mu_{i,+} = \frac{1}{2} + \varepsilon\theta_i, \qquad \mu_{i,-} = \frac{1}{2} - \varepsilon\theta_i, \qquad 0 < \varepsilon \le \frac{1}{4}. \tag{25}$$

For a selected k-set S, put

$$a_i(S) = \mathbf{1}\{(i,+) \in S\} - \mathbf{1}\{(i,-) \in S\} \in \{-1, 0, 1\}.$$

Then

$$Y = \frac{k}{2} + \varepsilon\langle a(S), \theta\rangle + \xi, \qquad \xi \sim N(0, k), \tag{26}$$

and the unnormalized gap is

$$\Delta_\theta^{\mathrm{raw}}(S) = \varepsilon\{k - \langle a(S), \theta\rangle\}. \tag{27}$$

This representation covers every k-subset, including subsets that choose both arms from one pair and neither arm from another: $a(S) \in \{-1, 0, 1\}^k$ and $\|a(S)\|_2^2 \le k$. Under any posterior, one arm in each pair has posterior mean at least $1/2$ and the other at most $1/2$; hence a Bayes-optimal k-subset chooses the posterior-preferred arm from every pair. We apply the general theory to $\Delta_\theta = \Delta_\theta^{\mathrm{raw}}/k \in [0, 1]$.

Remark 3.2 (Dense unrestricted queries versus grouped queries). *For any $a \in \{-1,0,1\}^k$, choose $u, v \in \{-1,+1\}^k$ with $u + v = 2a$. These are two feasible one-arm-per-pair queries. If Y_u, Y_v are their independent aggregate observations, then*

$$\widetilde{Y} = \frac{Y_u + Y_v}{2} + \zeta, \qquad \zeta \sim N(0, k/2),$$

has exactly the law

$$\widetilde{Y} \sim N(k/2 + \varepsilon\langle a, \theta\rangle, k)$$

of the unrestricted query, while

$$\Delta_\theta^{\mathrm{raw}}(u) + \Delta_\theta^{\mathrm{raw}}(v) = 2\Delta_\theta^{\mathrm{raw}}(a).$$

Thus an unrestricted T-query policy can be simulated by a grouped $2T$-query policy with exactly twice its regret. This explains both the closeness to Audibert et al. (2014) and why our novelty claim is restricted to the exact stochastic model and stopped-cost formulation.

Let $m_i(H) = \mathbb{E}[\theta_i \mid H]$. The posterior risk is therefore exactly

$$r(Q_H) = \frac{\varepsilon}{k}\sum_{i=1}^{k}\{1 - |m_i(H)|\}. \tag{28}$$

In particular the product prior Q_0 is active at scale $b = \varepsilon$.

Proposition 3.3 (Stopped cost of the paired family). *For every fixed $0 < \gamma < 1$, uniformly over $0 < \varepsilon \le 1/4$,*

$$\mathsf{SC}_\varepsilon(Q_0) = \Theta_\gamma(k). \tag{29}$$

Proof architecture. For the upper stopped bound, a fixed Rademacher design of $N = O_\gamma(k/\varepsilon^2)$ feasible sign queries has a well-conditioned Gram matrix. Least squares then has constant coordinatewise sign error. The posterior Bayes sign rule can only improve on this estimator, so Markov's inequality gives constant-probability contraction of (28). The prior expected normalized gap of every predetermined query is ε, yielding stopped cost $O_\gamma(k)$.

For the reverse bound, the stopped direct-product lemma (Lemma B.1) gives, for every experiment with success probability p,

$$\mathbb{E}N \ge c_\gamma \frac{pk}{\varepsilon^2}. \tag{30}$$

A successful transcript has $\sum_i |m_i| \ge (1-\gamma)k$. Flipping coordinate i changes a conditional Gaussian mean by $2\varepsilon a_i$, while $\sum_i a_i^2 \le k$ cancels the aggregate noise variance k. Lemma 2.2 therefore turns (30) into stopped cost $\Omega_\gamma(k)$. The complete design, posterior-risk, and stopping-time arguments are in Appendix B.

What the stopped conversion contributes here. Constant-factor Hamming or Bayes-risk contraction on the paired family has sample complexity $\Theta(k/\varepsilon^2)$. That fact alone does not imply (24): a sample may be statistically informative while having very different decision loss, and a finite-horizon policy may achieve contraction only on a rare subset of transcripts. Proposition 3.3 couples information, regret, and the probability of posterior contraction for the same stopped policy. Theorem 2.3 then handles both the branch that contracts and the branch that continues to pay risk. This is precisely the step absent from a fixed-confidence sample-complexity argument. A direct horizon-wise Assouad proof, or the two-query reduction in Remark 3.2, can recover this particular lower bound. For this application, the modular contribution is therefore the following: once the model-specific stopped cost is known, the general two-branch theorem converts it into regret. Choosing $\varepsilon \asymp \sqrt{k/T}$ for $T \gtrsim k$, and a constant ε for $T \lesssim k$, gives the two regimes in Theorem 3.1.

4 Preference feedback on the paired family

We now apply the stopped-cost method to a concrete preference-feedback problem. Let $H \geq k$, and consider a deterministic H-stage chain. The first k states have actions $\{-1, +1\}$, and the remaining states have one action. Thus a deterministic policy is a sign vector $x \in \{-1, +1\}^k$. For $\theta \in \{-1, +1\}^k$, its latent value is

$$V_\theta(x) = \frac{H}{2} + \varepsilon\langle x, \theta\rangle, \qquad 0 < \varepsilon \leq \frac{1}{4}.$$

This is realized by rewards $1/2 + \varepsilon x_i \theta_i$ at the first k stages and $1/2$ thereafter. Transitions are known, but numerical rewards are not observed.

One preference query chooses two policies x, x', executes two episodes, and observes a binary label

$$\mathbb{P}_\theta(Y = 1 \mid x, x') = F\left(\beta \frac{V_\theta(x) - V_\theta(x')}{H}\right), \qquad F \in \{\Phi, \sigma\}, \tag{31}$$

where Φ is the standard Gaussian cdf and $\sigma(u) = (1 + e^{-u})^{-1}$. The factor β measures the evaluator's sensitivity to normalized value differences. The policy-pair macro-action has average normalized value gap

$$\Delta_\theta(x, x') = \frac{2V_\theta^\star - V_\theta(x) - V_\theta(x')}{2H}. \tag{32}$$

Hence the sum of the normalized regrets of the two executed episodes is exactly $2\Delta_\theta(x, x')$.

Let Q_0 be uniform on $\{-1, +1\}^k$, and put

$$\alpha = \frac{\beta\varepsilon}{H}, \qquad b = \frac{\varepsilon k}{H}.$$

If $m_i(h) = \mathbb{E}[\theta_i \mid h]$, minimization over the two policies separates:

$$r(Q_h) = \frac{\varepsilon}{H} \sum_{i=1}^{k} \{1 - |m_i(h)|\}, \qquad r(Q_0) = b. \tag{33}$$

This is the same posterior Hamming risk as in (28), but the observation channel is binary.

Proposition 4.1 (Stopped cost for paired trajectory preferences)**.** *Fix $0 < \gamma < 1$ and $F \in \{\Phi, \sigma\}$. Then*

$$\mathsf{SC}_b^{\mathrm{pref}}(Q_0) = \Theta_{\gamma,F}\left(b^2\{k + \alpha^{-2}\}\right) = \Theta_{\gamma,F}\left(\frac{\varepsilon^2 k^3}{H^2} + \frac{k^2}{\beta^2}\right). \tag{34}$$

Let

$$D_{k,H,\varepsilon,\beta} = k + \frac{H^2}{\beta^2\varepsilon^2}.$$

With $H, k, \varepsilon, \beta, F$ known to the learner, define the fixed-scale minimax pair-average normalized value regret by

$$\mathfrak{R}^{\mathrm{pref}}_{k,H,\varepsilon,\beta}(T) = \inf_{\pi} \sup_{\theta \in \{-1,+1\}^k} \mathbb{E}^{\pi}_{\theta} \sum_{t=1}^{T} \Delta_\theta(X_t, X'_t).$$

For $T \geq 2$, it satisfies

$$c_{\gamma,F}\, b \min\{T, D_{k,H,\varepsilon,\beta}\} \leq \mathfrak{R}^{\mathrm{pref}}_{k,H,\varepsilon,\beta}(T) \leq C_F\, b \min\{T, D_{k,H,\varepsilon,\beta} \log(eT)\}. \tag{35}$$

Calculation. For the upper stopped bound, compare policies that disagree with random signs on a random subset of

$$s \asymp \max\{1, \min(k, \alpha^{-2})\}$$

coordinates and agree randomly elsewhere. A signed-correlation estimator has effective coordinate-wise signal-to-noise ratio squared $\Omega_F((k + \alpha^{-2})^{-1})$. After $O_{\gamma,F}(k + \alpha^{-2})$ comparisons, its expected Hamming error is a sufficiently small fraction of k; the posterior Bayes sign rule can only improve on it. Every queried pair has prior expected gap b, so this proves the upper half of (34).

For the reverse inequality, if p is the contraction probability, a successful transcript has

$$\sum_i |m_i| \geq (1-\gamma)k, \qquad \sum_i m_i^2 \geq (1-\gamma)^2 k.$$

Conditioning on the policy's independent random seed, the stopped label strings form a prefix-free set, so mutual information gives $\mathbb{E}N \gtrsim_\gamma pk$. Coordinate-flip KL gives a second bound. Data processing for the probit channel and

$$D_{\mathrm{KL}}(\mathrm{Ber}\,\sigma(u) \| \mathrm{Ber}\,\sigma(v)) \leq \frac{1}{8}(u-v)^2$$

for the logistic channel imply

$$2^{-k} \sum_{\theta,i} D_{\mathrm{KL}}(P_\theta \| P_{\theta^{(i)}}) \lesssim_F \alpha^2 k\, \mathbb{E}N.$$

The same posterior-magnetization identity as in Lemma B.1 lower-bounds the left side by $\Omega_\gamma(pk)$, so $\mathbb{E}N \gtrsim_{\gamma,F} p\alpha^{-2}$. Before contraction the expected pair gap is at least γb, proving the stopped lower bound. The lower inequality in (35) is then the stopped conversion. A fixed-confidence correlation design followed by the estimated best pair gives the upper inequality. Appendix C supplies the complete argument.

Comparisons generated by noisy aggregate trajectory scores. Take $H = k$, and suppose the evaluator reports which of two independent latent aggregate scores is larger, where

$$G_\theta(x) \sim N(V_\theta(x), k).$$

Then

$$\mathbb{P}\{G_\theta(x) > G_\theta(x')\} = \Phi\left(\frac{\varepsilon \langle x - x', \theta \rangle}{\sqrt{2k}}\right),$$

which is (31) with $\beta = \sqrt{k/2}$. Consequently

$$D_{k,k,\varepsilon,\sqrt{k/2}} = \Theta(k/\varepsilon^2), \qquad \mathsf{SC}^{\mathrm{pref}}_{\varepsilon}(Q_0) = \Theta_\gamma(k),$$

and the raw pair-average value regret is

$$\widetilde{\Theta}\left(\min\{k\varepsilon T, k^2/\varepsilon\}\right). \tag{36}$$

Maximizing this fixed-scale rate over $0 < \varepsilon \le 1/4$ gives $\widetilde{\Theta}(\min\{kT, k^{3/2}\sqrt{T}\})$, matching the dense scaling in Theorem 3.1. For a fixed-slope Bradley–Terry evaluator on normalized values, the stopped cost is instead $\Theta_\gamma(k^2/\beta^2)$ when $\beta\varepsilon \lesssim \sqrt{k}$. Thus the rate depends explicitly on the evaluator's sensitivity. In every case one label consumes two episodes: total raw regret over the $2T$ executed episodes is twice (36).

Scope relative to preference-based RL. Xu et al. (2020) give finite-time PAC policy-identification guarantees and already connect trajectory preferences to full-bandit combinatorial feedback. General cumulative-regret results for linear trajectory preferences and reductions from preference to reward feedback are developed by Saha et al. (2023) and Wang et al. (2023). Proposition 4.1 contributes an exact model-specific stopped cost and value-regret calculation, including the quality of both queried trajectories and the hit probability. The chain has known deterministic dynamics, so its statistical core is a structured combinatorial dueling bandit; we do not claim a new general tabular-RL minimax rate. For RLHF and language-model preference learning, the formula makes two costs explicit: the sensitivity β of the binary evaluator and the value loss of both sampled responses. It does not model function approximation, changing evaluators, or off-policy generalization. Unknown transitions and reachability-dependent comparisons are the main unresolved RL extensions.

5 Gaussian feedback on compact action spaces

We next identify a broad class in which pointwise stopped experiments can be selected measurably as functions of the posterior and worst-prior Bayes regret coincides with frequentist minimax regret. Let $\mathcal{C}_d$ be a collection of nonempty compact metric action spaces. For $C \in \mathcal{C}_d$, let $\mathfrak{F}_d(C)$ be a nonempty Borel subset of $\mathcal{C}(C, [0,1])$, with the uniform topology. If f is fixed, querying $x \in C$ reveals

$$O = f(x) + \xi, \qquad \xi \sim N(0,1),$$

and incurs the gap $f(x) - \min_C f$. Neither C nor f is assumed to be convex in this section. Write

$$\mathfrak{R}^{\mathrm{M}}_{\mathfrak{F}}(d,T) = \sup_{C\in\mathcal{C}_d} \inf_{\pi} \sup_{f\in\mathfrak{F}_d(C)} \mathbb{E}^{\pi}_{f} R_T, \tag{37}$$

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{F}}(d,T) = \sup_{C\in\mathcal{C}_d} \sup_{Q \text{ on } \mathfrak{F}_d(C)} \inf_{\pi} \mathbb{E}^{\pi}_{Q} R_T. \tag{38}$$

Both infima are over Borel randomized nonanticipating policies. The values $\mathsf{SC}^{\mathrm{pt}}_{\mathfrak{F},C}$ and $\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{F},C}$ are defined by (9)–(10); an outer supremum over $C \in \mathcal{C}_d$ gives $\mathsf{SC}^{\mathrm{pt}}_{\mathfrak{F}}$ and $\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{F}}$.

Theorem 5.1 (Gaussian statewise selection and minimax equality)**.** *For every finite d, T, B,*

$$\mathfrak{R}^{\mathrm{M}}_{\mathfrak{F}}(d,T) = \mathfrak{R}^{\mathrm{B}}_{\mathfrak{F}}(d,T), \tag{39}$$

$$\mathsf{SC}^{\mathrm{pt}}_{\mathfrak{F},C}(d,B) = \mathsf{SC}^{\mathrm{unif}}_{\mathfrak{F},C}(d,B). \tag{40}$$

The supremum in (39) may be restricted to finitely supported priors. For every $\epsilon > 0$, the second equality is implemented by a universally measurable state-dependent experiment whose finite deterministic cap may depend on the state.

The first equality is a finite-horizon minimax theorem, not an asymptotic interchange. The second identifies the statewise and uniform stopped values. Convexity is not needed. Compactness and continuity ensure that Bayes actions exist; lower-semianalytic dynamic programming supplies universally measurable near-minimizing experiments; and the strictly positive Gaussian density makes the posterior update Borel. Write their common worst-space value as $\mathsf{SC}_{\mathfrak{F}}(d, B)$, and use the exponents from (16)–(18) with $\mathfrak{D} = \mathfrak{F}$.

Corollary 5.2 (Regret and stopped exponents under Gaussian feedback)**.** *For every sequence of the Gaussian loss models above,*

$$\alpha_{\mathfrak{F}}^{\mathrm{M}} = \alpha_{\mathfrak{F}}^{\mathrm{B}} = \frac{\alpha_{\mathfrak{F}}^{\mathrm{stop}}}{2}. \tag{41}$$

No information-scale matching or nondegeneracy assumption is required.

Proof. Apply Theorem 2.4 and (39)–(40). □

5.1 The full convex class

We now return to stochastic BCO: $C \subset \mathbb{R}^d$ is a compact convex body and $\mathfrak{F}_d(C)$ is the full class of continuous $[0, 1]$-valued convex losses. Convexity first becomes essential here, through the model class and the BCO bounds used below.

Corollary 5.3 (Full-class exponents)**.** *For the full class of continuous* $[0, 1]$*-valued convex losses,*

$$\frac{5}{2} \le \alpha_{\mathrm{full}}^{\mathrm{stop}} \le 3, \qquad \frac{5}{4} \le \alpha_{\mathrm{full}}^{\mathrm{reg}} \le \frac{3}{2}. \tag{42}$$

Determining either exponent determines the other.

The upper endpoint follows from the current general BCO upper bounds (Lattimore and György, 2023; Fokkema et al., 2024); the lower endpoint follows from Rajaraman (2026). Appendix G records the noise-preserving rescaling needed to place the latter hard family in the present model.

6 Observation-adaptive experiments: the dimple family

Let $C = B_2^d(1)$. Fix

$$d^{-1} \le \varepsilon \le \varepsilon_0, \qquad s = \sqrt{1 + 2\varepsilon},$$

where $\varepsilon_0 > 0$ is a sufficiently small constant. For $\theta \in S_2^{d-1}(1)$, define

$$F_\theta(x) = \varepsilon + \frac{1}{2}\|x\|^2 - \frac{1}{2}\big(s - \|\theta - x\|\big)_+^2. \tag{43}$$

These functions are continuous, convex, and $[0, 1]$-valued after reducing ε_0 if necessary (Bakhtiari et al., 2025, Lemma 10). Their unique minimizers are θ, with $F_\theta(\theta) = 0$. At the origin,

$$b_0 := F_\theta(0) = s - 1 = \frac{2\varepsilon}{\sqrt{1 + 2\varepsilon} + 1} \asymp \varepsilon. \tag{44}$$

Bakhtiari et al. (2025) use this family to exhibit an obstruction to standard one-step information-ratio analysis. We show that the same geometry admits a cheaper multistep experiment. The lower bound below uses mutual information to show that an unoriented ray reveals very little; the upper construction uses early observations to orient later tangent probes.

Use the rotation-invariant prior $\Theta \sim \mathrm{Unif}(S_2^{d-1}(1))$. Since $\nabla F_\theta(0) = -b_0\theta$, convexity gives

$$F_\theta(x) \geq b_0\{1 - \langle \theta, x\rangle\}.$$

The origin is therefore the prior Bayes action and $r(Q_0) = b_0$. Write

$$\tau_\gamma = \inf\{t \geq 1 : r(Q_t) \leq \gamma b_0\}. \tag{45}$$

Definition 6.1 (Fresh directions)**.** *A policy uses* fresh directions *if, at round t, it first draws $B_t \sim \mathrm{Unif}(S_2^{d-1}(1))$, independently of (Θ, H_{t-1}), then chooses an arbitrary $\sigma(H_{t-1}, B_t)$-measurable radius $R_t \in [0,1]$ and queries $X_t = R_tB_t$.*

The radius may use the complete past. Only the direction must be fresh. Every ray starts at the origin. The comparison excludes shifted or paired probes, anisotropic directions, and reuse of a learned direction.

Theorem 6.2 (Fresh-direction information bound)**.** *For $d \geq 2$, there is a universal $C < \infty$ such that, for every posterior ν of Θ, every fresh $B \sim \mathrm{Unif}(S_2^{d-1}(1))$, every measurable $R = R(B) \in [0,1]$, and independent $\xi \sim N(0,1)$,*

$$I_\nu(\Theta; B, F_\Theta(RB) + \xi) \leq C(\varepsilon^4 + d^{-4}). \tag{46}$$

Under the spherical prior, every nonadaptive batch $X_1, \ldots, X_n$, possibly randomized but independent of Θ and fixed before its observations, satisfies

$$I(\Theta; O_{1:n} \mid X_{1:n}) \leq Cn(\varepsilon^4 + d^{-4}). \tag{47}$$

For any initial prior, every stopped fresh-direction experiment with duration N and $\mathbb{E}N < \infty$ satisfies

$$I(\Theta; H_N) \leq C(\varepsilon^4 + d^{-4})\,\mathbb{E}N. \tag{48}$$

Conversely, under the spherical initial prior, if $\mathsf{M} \in \sigma(H_N)$ is an event on which the terminal posterior Bayes risk is at most γb_0, then

$$I(\Theta; H_N) \geq \frac{d-1}{2}(1-\gamma)^2\,\mathbb{P}(\mathsf{M}). \tag{49}$$

Consequently, every stopped fresh-direction experiment under the spherical prior with $\mathbb{P}(\mathsf{M}) > 0$ satisfies

$$\frac{\mathbb{E}N}{\mathbb{P}(\mathsf{M})} \geq c_\gamma \frac{d}{\varepsilon^4 + d^{-4}}. \tag{50}$$

Under the spherical prior, the same bound holds with $N = n$ for an arbitrary nonadaptive batch of size n. Thus, under that prior, constant-probability contraction requires batch size, or expected fresh-direction duration, $\Omega_\gamma(d/(\varepsilon^4 + d^{-4}))$.

Fix a sufficiently large $C_L = C_L(\gamma)$, and for $0 < a \leq b_0$ put

$$L_{d,a} = 1 + \log\left\{\frac{C_L d(1 + \log(d/a))}{a}\right\}. \tag{51}$$

Theorem 6.3 (Observation-adaptive directional experiment)**.** *There are constants $\gamma_0, c_0 > 0$, with $\gamma_0 + c_0 < \gamma$, and $C_\gamma < \infty$ such that, for every sufficiently large d, every $d^{-1} \le \varepsilon \le \varepsilon_0$, and every $0 < a \le b_0$, an observation-adaptive experiment returns $\widehat{x} \in B_2^d(1)$ with*

$$\sup_{\theta \in S_2^{d-1}(1)} \mathbb{P}_\theta\{F_\theta(\widehat{x}) > \gamma_0 a\} \le c_0 a, \tag{52}$$

$$N \le \frac{C_\gamma d^2 L_{d,a}^2}{a^2}, \tag{53}$$

$$\sup_{\theta \in S_2^{d-1}(1)} \mathbb{E}_\theta R_N \le \frac{C_\gamma d^2 L_{d,a}^2}{a}. \tag{54}$$

The query cap is deterministic. The conditional distributions and subspaces of the directions in the last two stages depend measurably on earlier observations. Under every prior on Θ,

$$\mathbb{P}\left\{\mathbb{E}[F_\Theta(\widehat{x}) \mid H_N] \le \gamma a\right\} \ge 1 - \frac{\gamma_0 + c_0}{\gamma} > 0. \tag{55}$$

The construction has three stages. The table suppresses logarithmic factors.

stage	progress	(queries, regret)
orthogonal scan	obtain v_0 with $\langle \theta, v_0 \rangle = \widetilde{\Omega}(\max\{\sqrt{\varepsilon}, d^{-1/2}\})$	$(d^2/\varepsilon^2,\ d^2/\varepsilon)$
tangent	choose tangent directions from the current estimate	$(d^2/\varepsilon^2,\ d^2/\varepsilon)$
amplification	and enter a fixed angular cone	
angular contraction	invert boundary observations and repeat $q \mapsto q/2$	$(d^2/a^2,\ d^2/a)$
	until $q = O(a)$	

Only the last two stages orient later probes using earlier observations. Their detailed analysis is in Appendix F.

For the next statement, $\mathsf{SC}_b^{[1]}$ restricts (5) to one query, and $\mathsf{SC}_b^{\text{fresh}}$ restricts it to fresh-direction policies.

Corollary 6.4 (Polynomial separations)**.** *For all sufficiently large d, at $\varepsilon = d^{-1}$ and $b = b_0 \asymp d^{-1}$, the initial spherical state satisfies*

$$\mathsf{SC}_b^{[1]} = \Omega_\gamma(d^3), \qquad \mathsf{SC}_b^{\text{fresh}} = \Omega_\gamma(d^3), \qquad \mathsf{SC}_b = \widetilde{O}_\gamma(d^2). \tag{56}$$

At the same scale and under the spherical prior, a nonadaptive batch with constant contraction probability has size $\Omega_\gamma(d^5)$, and every stopped fresh-direction experiment with $\mathbb{P}(\mathsf{M}) > 0$ satisfies

$$\frac{\mathbb{E}N}{\mathbb{P}(\mathsf{M})} \ge c_\gamma d^5.$$

In particular, constant success probability requires $\mathbb{E}N = \Omega_\gamma(d^5)$. The observation-adaptive construction contracts risk with constant probability using $\widetilde{O}_\gamma(d^4)$ queries.

Let

$$L_{d,T} = 1 + \log\left\{C_L d \max\{T, d\}[1 + \log\{d \max\{T, d\}\}]\right\}.$$

Under the spherical prior, every fresh-direction policy satisfies

$$\mathbb{E}R_T \ge c_\gamma \min\left\{\frac{T}{d}, d^4\right\}, \tag{57}$$

while an unrestricted policy satisfies, uniformly in θ,

$$\mathbb{E}_\theta R_T \le C_\gamma L_{d,T} \min\left\{\frac{T}{d}, d\sqrt{T}\right\}. \tag{58}$$

At $T = d^5$, these bounds are $\Omega(d^4)$ and $\widetilde{O}(d^{7/2})$, respectively.

The stopped-cost comparison follows by combining the two information bounds in Theorem 6.2 with Lemma 2.2, and by truncating the adaptive construction at its first hit. The regret bounds then use the same two-branch argument as (15) and an explore–then–commit use of Theorem 6.3. Full details are in Appendices D and E.

For the full BCO class, closing $\alpha_{\text{full}}^{\text{stop}} \in [5/2, 3]$ requires either uniformly low-cost posterior experiments or a lower bound that rules out all observation-adaptive probes. The dimple result explains why fixed batches and fresh directions cannot decide that question.

7 Further scope and open problems

7.1 General episodic reinforcement learning

Example 7.1 (Episodic reinforcement learning)**.** *Let M be a fixed unknown MDP with episode length H, rewards in $[0,1]$, and a common initial-state distribution ρ. One macro-action is a complete contingent policy π for an episode, and its observation is the resulting trajectory. After normalizing by H, put*

$$\Delta_M(\pi) = \frac{V_M^\star(\rho) - V_M^\pi(\rho)}{H} \in [0,1].$$

At episode boundaries,

$$r(Q) = \inf_\pi \mathbb{E}_Q \Delta_M(\pi)$$

satisfies (2), and stopping occurs between episodes. Thus the framework measures episodic pseudo-regret; a realized reward difference need not itself be nonnegative.

The reset to ρ is essential. In continuing control, the physical state changes and $r(Q_t, S_t)$ need not be a supermartingale: the next state can be intrinsically harder without resolving uncertainty. Such problems need a risk-persistence condition or a stateful Bellman formulation such as Xu (2026a).

Corollary 7.2 (Finite episodic models)**.** *Consider any sequence of reset episodic reinforcement-learning problems as in Example 7.1, with a finite model class, finite state, action, and reward alphabets, and a finite episode horizon. Using the normalized episode gap,*

$$\alpha_{\text{RL}}^{\text{M}} = \alpha_{\text{RL}}^{\text{B}} = \frac{\alpha_{\text{RL}}^{\text{stop}}}{2}. \tag{59}$$

For fixed H, normalization does not change the dimension exponent. If H grows with d, its dependence must be restored in unnormalized regret.

For preference feedback, the macro-action is a policy pair and the gap is the average normalized value loss in (32). Section 4 calculates this object for one structured known-dynamics chain. In a general finite model, the joint latent class of the MDP and comparison kernel must be finite (or satisfy separate compactness and selection conditions). A changing evaluator, deployment in a different MDP, or unknown transitions with reachability constraints requires additional state and a new model-specific stopped experiment.

7.2 Extreme contextual recommendation

Top-k contextual bandits are used when the arm set is extremely large. Sen et al. (2021) use an arm hierarchy to reduce millions of arms to a context-dependent candidate set, with experiments motivated by advertising and recommendation. Their feedback may reveal all or some selected-arm rewards, whereas Theorem 3.1 assumes only the aggregate and therefore does not lower-bound their algorithm.

The connection instead exposes two distinct scales. Arm hierarchies make search over a large catalog computationally feasible; within the localized candidate set, loss of individual attribution can still carry a k-dependent statistical price. A contextual stopped profile would condition on context and posterior, charge aggregate probes, and stop when context-averaged top-k risk contracts. The dense theorem proves that the attribution price can be real, but it does not transfer automatically to semi-bandit feedback or realizable contextual regression.

Across the applications, the division of labor is constant. The model-specific step is local: construct or rule out a regret-efficient posterior-contraction experiment at one scale. The model-independent transfer then concatenates contractions for upper bounds and stops any regret policy at its first contraction for lower bounds. After checking posterior sufficiency, risk persistence, measurability, and Bayes–minimax closure when needed, a new model requires a new experimental-design argument, not a new regret-conversion proof.

A Proofs for the stopped conversion

Proof of Lemma 2.2. The event $\{t \leq N\}$ lies in $\mathcal{H}_{t-1}$, and on this event the pre-query posterior has not hit the lower-risk set. If π_t is the randomized action kernel, then

$$\mathbb{E}[\Delta_\omega(A_t) \mid \mathcal{H}_{t-1}] = \int \mathbb{E}_{Q_{t-1}} \Delta_\omega(a)\, \pi_t(da \mid H_{t-1}) \geq r(Q_{t-1}) > \gamma b.$$

Tonelli's theorem applied to the stopped sum proves (6); the comparison follows immediately. $\square$

Proof of Theorem 2.3. If $\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d,T) = +\infty$, the upper statement is the trivial bound T. Otherwise fix one universally measurable selector Σ whose worst-state ratio is at most $\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d,T) + \epsilon$. For the upper bound, use geometric risk buckets $b_j = \vartheta^j$. Whenever $r(Q_t) > 2\eta$, invoke $\Sigma(Q_t, b_j)$ at the unique active scale; otherwise play a measurable δ_t-Bayes action, where $\sum_t \delta_t \leq \delta$. If the horizon cuts through the final experiment block, extend that block only for the purpose of the analysis until its prescribed stopping time; the actual online policy still stops at T. Because every block makes at least one actual query, at most one block is completed in this virtual manner. At scale b_j, the selector's ratio bound gives

$$b_j\, \mathbb{E}[R_k \mid H_{S_k}] \leq \{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d,T) + \epsilon\} p_k,$$

where p_k is the conditional hit probability. If $\mathcal{K}_j$ is the random set of blocks invoked in bucket j and M_k is the hit indicator for block k, conditional expectation gives the accounting identity

$$\mathbb{E} \sum_{k \in \mathcal{K}_j} p_k = \mathbb{E} \sum_{k \in \mathcal{K}_j} M_k = \mathbb{E}[\text{number of actual hits in bucket } j]. \tag{60}$$

Thus unsuccessful blocks enter the regret sum through their conditional hit probabilities but do not create an additional term.

After the k-th hit in bucket j, at time U_k, one has $r(Q_{U_k}) \leq \gamma b_j$. By (2) and Doob's maximal inequality, applied first over finite horizons after U_k and then by monotone convergence,

$$\mathbb{P}\left\{ \sup_{t \geq U_k} r(Q_t) > \vartheta b_j \,\middle|\, \mathcal{H}_{U_k} \right\} \leq \frac{\gamma}{\vartheta}.$$

A later return to the same active bucket requires the event inside this probability. The number of hits in one bucket is therefore geometrically dominated and has expectation at most $(1-\gamma/\vartheta)^{-1}$. Hence

$$\mathbb{E} R_{\mathrm{explore}} \leq \frac{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d,T) + \epsilon}{1-\gamma/\vartheta} \sum_{b_j \geq 2\eta} \frac{1}{b_j} \leq \frac{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d,T) + \epsilon}{2(1-\vartheta)(1-\gamma/\vartheta)\eta}.$$

The non-experiment rounds contribute at most $2T\eta + \delta$. Letting $\delta, \epsilon \downarrow 0$ proves (11); optimizing η, with the boundary choices and the trivial bound T, proves (12).

For the converse, choose an active state with $\mathsf{SC}_b(Q) > c$. Run an arbitrary n-round policy, stop it at $N = \tau_b \wedge n$, and put $p = \mathbb{P}_Q(\tau_b \leq n)$. The stopped-cost premise and nonnegativity give

$$\mathbb{E}_Q R_n \geq \mathbb{E}_Q R_N \geq \frac{c}{b}\, p.$$

By the same pre-hit argument as in Lemma 2.2,

$$\mathbb{E}_Q R_n \geq \mathbb{E}_Q R_N \geq \gamma b\, \mathbb{E}_Q N \geq \gamma b n(1-p).$$

Minimizing the maximum of these two affine functions over $p \in [0,1]$ gives (13). At the displayed calibrated horizon, $n \geq c/(2\gamma b^2)$, and direct substitution gives $\mathbb{E}_Q R_n \geq c/(4b) \geq (\sqrt{\gamma}/4)\sqrt{cn}$. $\square$

Proof of Theorem 2.4. For $T \le d^M$, Theorem 2.3 and monotonicity of the scale window give

$$\frac{\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d,T)}{\sqrt{T}} \le C_{\gamma,\vartheta}\left\{T^{-1/2} + \sqrt{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d,d^M)}\right\}.$$

Taking exponents proves the upper inequality in (19).

For the reverse inequality, first suppose $\alpha^{\mathrm{stop,pt}}_{\mathfrak{D}} > 0$, and fix

$$0 < q < q' < \alpha^{\mathrm{stop,pt}}_{\mathfrak{D}}.$$

For some fixed m and an infinite sequence of dimensions,

$$1 + \mathsf{SC}^{\mathrm{pt}}_{\mathfrak{D}}(d,d^m) \ge d^{q'}.$$

For all sufficiently large dimensions in this sequence, there is an active state (Q_d, b_d), with $b_d \ge d^{-m}$, such that

$$\mathsf{SC}_{b_d}(Q_d) > c_d, \qquad c_d = d^q.$$

Set

$$n_d = \left\lfloor \frac{c_d}{\gamma b_d^2} \right\rfloor.$$

Then $c_d/(\gamma b_d^2) \ge 2$ for large d, and

$$n_d \le \gamma^{-1} d^{q+2m} \le d^M$$

for one fixed integer M. The calibrated lower bound in Theorem 2.3 yields

$$\frac{\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d,n_d)}{\sqrt{n_d}} \ge \frac{\sqrt{\gamma}}{4}\sqrt{c_d} = \frac{\sqrt{\gamma}}{4} d^{q/2}.$$

Thus $\alpha^{\mathrm{B}}_{\mathfrak{D}} \ge q/2$. Let $q \uparrow \alpha^{\mathrm{stop,pt}}_{\mathfrak{D}}$; if the latter exponent is infinite, take arbitrary finite q. When it is zero, the lower inequality is immediate.

The measurable-choice assumption identifies the two finite stopped values, and hence their exponents. Bayes–minimax equality identifies the remaining regret exponents. □

Proof of Corollary 2.5. Write $m = |\Omega_d|$. For each $a \in \mathcal{A}_d$, the measure

$$\lambda_a = \frac{1}{m} \sum_{\omega \in \Omega_d} P_\omega(\cdot \mid a)$$

dominates all $P_\omega(\cdot \mid a)$. Choose Borel Radon–Nikodym derivatives $p_{\omega,a}$. Off a predictive-null set, the posterior update is

$$\Psi(Q,a,o)(\omega) = \frac{Q(\omega) p_{\omega,a}(o)}{\sum_{\omega'} Q(\omega') p_{\omega',a}(o)},$$

with an arbitrary fixed value when the denominator is zero. Hence the belief transition is Borel. Moreover,

$$r(Q) = \min_{a \in \mathcal{A}_d} \sum_{\omega \in \Omega_d} Q(\omega) \Delta_\omega(a)$$

is Borel, and deterministic tie-breaking selects a Bayes action measurably.

For every finite cap L, the penalized stopped Bellman recursion (7) therefore has Borel primitives and a finite action set. Finite-horizon measurable selection supplies a measurable near-minimizer.

The rational-price argument in (8) then gives a universally measurable ϵ-optimal selector for the capped ratio. Finally, choose the first cap L whose value lies within $\epsilon/2$ of

$$\mathsf{SC}_b(Q) = \inf_{L\geq 1} \mathsf{SC}_{b,L}(Q).$$

This gives an ϵ-optimal selector for the uncapped stopped cost, so the pointwise and uniform profiles coincide.

For Bayes–minimax equality, let

$$\mathcal{V}_T = \left\{ \left(\mathbb{E}_\omega^\pi R_T \right)_{\omega\in\Omega_d} : \pi \right\} \subset [0,T]^m.$$

Independent randomization at H_0 makes $\mathcal{V}_T$ convex. The prior simplex $\mathcal{P}(\Omega_d)$ is compact, so Sion's minimax theorem gives

$$\inf_{v\in\mathcal{V}_T} \max_{\omega} v_\omega = \sup_{Q\in\mathcal{P}(\Omega_d)} \inf_{v\in\mathcal{V}_T} \langle Q, v\rangle.$$

The two sides are frequentist minimax and worst-prior Bayes regret, respectively. The exponent identity now follows from Theorem 2.4. □

Proof of Corollary 7.2. The latent MDP class and the complete contingent-policy set are finite, as is the trajectory observation space. Apply Corollary 2.5 to the episode-level problem. □

B Proofs for aggregate-feedback top-k bandits

Proof of Proposition 3.3. We first prove the upper bound for arbitrary k, without assuming the existence of a Hadamard matrix. For

$$n_0 = \left\lceil C_\gamma k/\varepsilon^2 \right\rceil,$$

there is a deterministic matrix $A \in \{-1,+1\}^{n_0\times k}$ such that

$$A^\top A \succeq \frac{n_0}{2} I_k. \tag{61}$$

Indeed, take independent Rademacher rows. For a fixed unit vector u, $\langle A_t, u\rangle$ is isotropic sub-Gaussian, so its empirical second moment differs from one by at most $1/4$ with probability $1-2e^{-cn_0}$. A $1/8$-net of the unit sphere has size at most 17^k; a union bound followed by the standard net extension gives (61) when $n_0 \geq Ck$. Increasing C_γ covers all $0<\varepsilon\leq 1/4$.

Every row a_t is a feasible top-k query: choose $(i,+)$ when $a_{t,i}=1$ and $(i,-)$ when $a_{t,i}=-1$. Centering the observations in (26) gives

$$Z = \varepsilon A\theta + \xi, \qquad \xi \sim N(0, kI_{n_0}).$$

The least-squares estimator

$$\widehat{v} = \frac{1}{\varepsilon}(A^\top A)^{-1}A^\top Z$$

satisfies

$$\widehat{v} - \theta \sim N\left(0, \frac{k}{\varepsilon^2}(A^\top A)^{-1}\right), \qquad \mathrm{Var}(\widehat{v}_i - \theta_i) \leq \frac{2k}{\varepsilon^2 n_0} \leq \frac{2}{C_\gamma}.$$

Let $\widehat{\theta}_i = \mathrm{sign}(\widehat{v}_i)$, with an arbitrary convention at zero. A Gaussian tail bound gives

$$\mathbb{P}(\widehat{\theta}_i \neq \theta_i) \leq e^{-C_\gamma/4}.$$

Choose C_γ so that the right-hand side is at most $\gamma/16$.

Conditional on the transcript, the coordinatewise posterior sign rule has Hamming risk

$$\frac{1}{2}\sum_{i=1}^{k}(1-|m_i|),$$

and is no worse than any transcript-measurable estimator. Consequently,

$$\mathbb{E}\sum_{i=1}^{k}(1-|m_i|) \le 2\mathbb{E}d_{\mathrm{H}}(\widehat{\theta},\theta) \le \frac{\gamma k}{8}.$$

Markov's inequality shows that the terminal posterior risk in (28) is at most $\gamma\varepsilon$ with probability at least $7/8$.

Run this predetermined design, but stop at the first contraction. Since all gaps are nonnegative, first-hit truncation can only reduce its regret. For every predetermined row a_t, symmetry of the product prior gives

$$\mathbb{E}\Delta_\theta(S_t) = \varepsilon\left\{1 - \frac{1}{k}\mathbb{E}\langle a_t,\theta\rangle\right\} = \varepsilon.$$

Therefore, with $p \ge 7/8$,

$$\frac{\varepsilon\,\mathbb{E}R_N}{p} \le \frac{8}{7}\,\varepsilon^2 n_0 \le C_\gamma k,$$

after enlarging the constant to cover the ceiling and $k=1$.

For the reverse inequality, Lemma B.1 applied to (26) gives, for every admissible experiment with success probability p,

$$\mathbb{E}N \ge c_\gamma \frac{pk}{\varepsilon^2}.$$

Lemma 2.2 now yields

$$\frac{\varepsilon\,\mathbb{E}R_N}{p} \ge \gamma\varepsilon^2\frac{\mathbb{E}N}{p} \ge c_\gamma k.$$

□

Proof of Theorem 3.1. The upper bound is the minimum of the trivial kT bound and the distribution-free full-bandit guarantee $O(k\sqrt{nT}\log(eT))$ of Rejwan and Mansour (2020), specialized to $n=2k$. Their guarantee is stated for independent 1-sub-Gaussian arms, which includes the present unit-variance Gaussian model.

For the lower bound, fix a constant $0 < c_0 < c_\gamma$, where $c_\gamma k$ is the lower stopped cost in Proposition 3.3. If $T \ge Ck$, choose $\varepsilon = c\sqrt{k/T}$ with $c>0$ small enough that $\varepsilon \le 1/4$. Apply Theorem 2.3 with $B \ge \varepsilon^{-1}$, the paired prior Q_0, and stopped-cost threshold $c_0 k$. Equation (13) gives

$$\sup_Q \inf_\pi \mathbb{E}_Q R_T \ge \frac{1}{2}\min\left\{\gamma\varepsilon T, \frac{c_0 k}{\varepsilon}\right\} \ge c'\sqrt{kT}$$

in normalized units. The frequentist minimax value dominates this Bayes value, and multiplication by k restores sum-reward units.

If $T < Ck$, take a sufficiently small constant $\varepsilon \le 1/4$. The same display gives $\Omega(T)$ normalized regret (with constants adjusted on the overlap between the two ranges), and hence $\Omega(kT)$ in raw units. The case $T=1$ is included. □

B.1 A stopped direct-product lemma

Lemma B.1 (Stopped Assouad bound for Gaussian sums)**.** *Let θ be uniform on $\{-1,+1\}^k$. At time t, an adaptive policy chooses $a_t \in \{-1,0,1\}^k$, with $\|a_t\|_2^2 \le k$, and observes*

$$Y_t = \varepsilon\langle a_t, \theta\rangle + \xi_t, \qquad \xi_t \overset{\text{iid}}{\sim} N(0,k).$$

Let N be the first time that $\sum_i (1 - |\mathbb{E}[\theta_i \mid H_N]|) \le \gamma k$, or an earlier voluntary stopping time, and let M be the first alternative. If $p = \mathbb{P}(\mathsf{M})$, then

$$\mathbb{E}N \ge c_\gamma \frac{pk}{\varepsilon^2}.$$

Proof. For $i \in [k]$, pair each parameter θ with $\theta^{(i)}$, obtained by flipping its i-th coordinate, and run the two transcript laws with the same policy kernel. The stopped Gaussian chain rule gives

$$D_{\mathrm{KL}}(P_\theta^N \| P_{\theta^{(i)}}^N) = \frac{2\varepsilon^2}{k}\mathbb{E}_\theta \sum_{t \le N} a_{t,i}^2. \tag{62}$$

If $\mathbb{E}N = \infty$, the desired duration lower bound is immediate. Otherwise the identity follows first for $N \wedge L$, then from convergence of relative entropy along the increasing stopped-history σ-fields and Tonelli's theorem for the nonnegative conditional KL increments. Averaging over θ, summing over i, and using $\sum_i a_{t,i}^2 \le k$ yields

$$2^{-k}\sum_\theta \sum_{i=1}^k D_{\mathrm{KL}}(P_\theta^N \| P_{\theta^{(i)}}^N) \le 2\varepsilon^2 \mathbb{E}N. \tag{63}$$

For each i, let P_i^+ and P_i^- be the transcript laws conditional on $\theta_i = +1$ and $\theta_i = -1$, respectively. Joint convexity of relative entropy and the pairing above give

$$\sum_i \{D_{\mathrm{KL}}(P_i^+ \| P_i^-) + D_{\mathrm{KL}}(P_i^- \| P_i^+)\} \le 2^{1-k}\sum_\theta \sum_i D_{\mathrm{KL}}(P_\theta^N \| P_{\theta^{(i)}}^N).$$

Let $P = (P_i^+ + P_i^-)/2$, which is the same unconditional transcript law for every i. Bayes' rule gives

$$m_i(h) = \frac{dP_i^+ - dP_i^-}{dP_i^+ + dP_i^-}(h).$$

Consequently the i-th Jeffreys divergence is

$$D_{\mathrm{KL}}(P_i^+ \| P_i^-) + D_{\mathrm{KL}}(P_i^- \| P_i^+) = 4\mathbb{E}_P[m_i(H_N) \operatorname{arctanh} m_i(H_N)].$$

Since $x \operatorname{arctanh} x \ge x^2$, if a stopped-transcript event M satisfies

$$\sum_i (1 - |\mathbb{E}[\theta_i \mid H_N]|) \le \gamma k \quad \text{on } M,$$

then Cauchy–Schwarz yields, pointwise on M,

$$\sum_i m_i(H_N)^2 \ge k^{-1}\left(\sum_i |m_i(H_N)|\right)^2 \ge (1-\gamma)^2 k.$$

It follows that

$$2^{-k}\sum_\theta \sum_{i=1}^k D_{\mathrm{KL}}(P_\theta^N \| P_{\theta^{(i)}}^N) \ge 2(1-\gamma)^2 k\, \mathbb{P}(M). \tag{64}$$

Take $M = \mathsf{M}$ and combine (63)–(64). ∎

C Proofs for paired trajectory preferences

Proof of Proposition 4.1. Write

$$\ell_F(u) = 2F(u) - 1.$$

For both $F = \Phi$ and $F = \sigma$, this is an odd increasing function. There are constants $c_F, C_F > 0$ such that the following correlation bound holds. For every $s \in [k]$ and $\alpha > 0$, define

$$\rho_s(\alpha) = \mathbb{E}\left[W_1 \ell_F\left(2\alpha \sum_{j=1}^{s} W_j\right)\right], \qquad W_1, \ldots, W_s \overset{\text{iid}}{\sim} \text{Unif}\{-1, +1\}.$$

Choose an integer

$$s_\alpha \asymp \max\{1, \min(k, \alpha^{-2})\}, \tag{65}$$

with the implicit constants fixed below. Then

$$\frac{s_\alpha}{k} \rho_{s_\alpha}(\alpha)^2 \geq \frac{c_F}{k + \alpha^{-2}}. \tag{66}$$

To verify this, condition on $S = \sum_{j=2}^{s} W_j$ and pair the two values of W_1:

$$\rho_s(\alpha) = \frac{1}{2}\mathbb{E}\left[\ell_F\{2\alpha(S+1)\} - \ell_F\{2\alpha(S-1)\}\right]. \tag{67}$$

If $\alpha^2 s \leq c_0$ for a sufficiently small absolute constant c_0, the event $|S| \leq 2\sqrt{s}$ has probability bounded below by an absolute constant. On this event the arguments between the two endpoints in (67) remain in a fixed compact interval. Both the Gaussian and logistic densities are bounded below there, and the mean-value theorem gives $\rho_s(\alpha) \geq c_F \alpha$. If α is bounded below and $s = 1$, then $\rho_1(\alpha) = \ell_F(2\alpha) \geq c_F$. Taking $s = k$ when $\alpha^2 k \leq c_0$, $s \asymp \alpha^{-2}$ when $c_0/k < \alpha^2 < c_1$, and $s = 1$ when $\alpha^2 \geq c_1$, proves (66).

Upper stopped bound. At each query, draw I uniformly from the s_α-subsets of $[k]$. For $i \in I$, draw ξ_i uniformly from $\{-1, +1\}$ and set

$$x_i = \xi_i, \qquad x_i' = -\xi_i.$$

For $i \notin I$, draw a common Rademacher sign and set $x_i = x_i'$ equal to that sign. If $L = 2Y - 1$, define

$$Z_i = \mathbf{1}\{i \in I\}\xi_i L.$$

By symmetry, for every fixed θ,

$$\mathbb{E}_\theta Z_i = \frac{s_\alpha}{k}\rho_{s_\alpha}(\alpha)\theta_i, \qquad \mathbb{E}_\theta Z_i^2 = \frac{s_\alpha}{k}. \tag{68}$$

The observations are independent across the predetermined randomized queries. Bernstein's inequality and (66) show that the estimator

$$\widehat{\theta}_i = \text{sign}\left(\sum_{t=1}^{n_0} Z_{t,i}\right)$$

satisfies

$$\mathbb{P}(\widehat{\theta}_i \neq \theta_i) \leq \exp\left\{-c_F \frac{n_0}{k + \alpha^{-2}}\right\}.$$

Choose

$$n_0 = \lceil C_{\gamma,F}(k+\alpha^{-2})\rceil$$

so that the last display is at most $\gamma/32$.

Conditional on the transcript, the coordinatewise posterior sign rule has Hamming risk

$$\frac{1}{2}\sum_{i=1}^{k}(1-|m_i|),$$

and is no worse than $\widehat{\theta}$. Therefore

$$\mathbb{E}\sum_{i=1}^{k}(1-|m_i|) \le 2\mathbb{E}d_{\mathrm{H}}(\widehat{\theta},\theta) \le \frac{\gamma k}{16}.$$

Markov's inequality implies

$$\mathbb{P}\{r(Q_{n_0}) \le \gamma b\} \ge \frac{15}{16}.$$

Run the design only until its first hit. Every complete predetermined query has prior expected gap

$$\mathbb{E}_{Q_0}\Delta_\theta(x,x') = \frac{\varepsilon k}{H} - \frac{\varepsilon}{2H}\mathbb{E}_{Q_0}\langle x+x',\theta\rangle = b.$$

First-hit truncation only reduces nonnegative regret. Hence

$$\frac{b\,\mathbb{E}R_N}{\mathbb{P}(\mathsf{M})} \le \frac{16}{15}b^2 n_0 \le C_{\gamma,F}b^2(k+\alpha^{-2}).$$

One-bit duration lower bound. Let an arbitrary stopped experiment have contraction event M and $p=\mathbb{P}(\mathsf{M})$. Let U be the policy's internal random seed, included in H_0 and independent of θ, and let $L_N=(Y_1,\ldots,Y_N)$ be the stopped label string. Conditional on U, the queried pairs and stopping decisions are functions of the preceding labels, and the possible strings L_N form a prefix-free set. The source-coding inequality and independence of U therefore give

$$I(\theta;\mathcal{H}_N) = I(\theta;L_N \mid U) \le (\log 2)\mathbb{E}N. \tag{69}$$

Since the prior is a product of unbiased signs, the chain rule for relative entropy and nonnegativity of posterior total correlation give

$$\begin{aligned} I(\theta;\mathcal{H}_N) &= \mathbb{E}D_{\mathrm{KL}}(Q_{\mathcal{H}_N}\|Q_0) \\ &\ge \sum_{i=1}^{k}\mathbb{E}\ \mathrm{kl}\left(\frac{1+m_i(\mathcal{H}_N)}{2}\,\middle\|\,\frac{1}{2}\right) \ge \frac{1}{2}\mathbb{E}\sum_{i=1}^{k}m_i(\mathcal{H}_N)^2. \end{aligned} \tag{70}$$

On M,

$$\sum_i |m_i| \ge (1-\gamma)k, \qquad \sum_i m_i^2 \ge (1-\gamma)^2 k.$$

Combining this with (69)–(70) yields

$$\mathbb{E}N \ge c_\gamma p k. \tag{71}$$

Link-slope duration lower bound. For $\theta^{(i)}$ obtained by flipping coordinate i, let

$$u = \alpha\langle x - x', \theta\rangle, \qquad v = \alpha\langle x - x', \theta^{(i)}\rangle.$$

Then

$$|u - v| = 2\alpha|x_i - x_i'|.$$

For the logistic link, the Bernoulli family in its natural parameter has log-partition curvature at most $1/4$, and hence

$$D_{\mathrm{KL}}(\mathrm{Ber}\,\sigma(u)\|\,\mathrm{Ber}\,\sigma(v)) \le \frac{1}{8}(u - v)^2.$$

For the probit link, thresholding $N(u, 1)$ at zero produces $\mathrm{Ber}\,\Phi(u)$, so data processing and the Gaussian KL formula give

$$D_{\mathrm{KL}}(\mathrm{Ber}\,\Phi(u)\|\,\mathrm{Ber}\,\Phi(v)) \le \frac{1}{2}(u - v)^2.$$

Thus, for either link, the stopped chain rule gives

$$2^{-k}\sum_{\theta}\sum_{i=1}^{k} D_{\mathrm{KL}}(P_\theta^N\|P_{\theta^{(i)}}^N) \le C_F\alpha^2 k\,\mathbb{E}N, \tag{72}$$

because $\sum_i (x_i - x_i')^2 \le 4k$ at every query. As in the proof of Lemma B.1, first establish the identity for $N \wedge L$ and pass to the limit; if $\mathbb{E}N = \infty$, the desired bound is immediate.

The joint-convexity and Jeffreys-divergence part of that proof depends only on the paired prior, not on the Gaussian observation law. It therefore applies unchanged and lower-bounds the left side of (72) by

$$2(1 - \gamma)^2 kp.$$

Consequently

$$\mathbb{E}N \ge c_{\gamma,F}\,p\alpha^{-2}. \tag{73}$$

Equations (71) and (73), together with $\max\{u, v\} \ge (u + v)/2$, imply

$$\mathbb{E}N \ge c_{\gamma,F}\,p(k + \alpha^{-2}).$$

Lemma 2.2 now yields

$$\frac{b\,\mathbb{E}R_N}{p} \ge \gamma b^2\frac{\mathbb{E}N}{p} \ge c_{\gamma,F}b^2(k + \alpha^{-2}),$$

which proves the stopped lower bound.

Fixed-scale regret bounds. Apply Theorem 2.3 at Q_0, with a constant strictly below the stopped lower bound just proved. This gives

$$\mathfrak{R}^{\mathrm{pref}}_{k,H,\varepsilon,\beta}(T) \ge c_{\gamma,F}\,b\min\{T, k + \alpha^{-2}\}.$$

For the reverse inequality, take

$$L_0 = \left\lceil C_F(k + \alpha^{-2})\log(eT)\right\rceil.$$

If $L_0 \ge T$, play arbitrary pairs; their gap is at most $2b$. If $L_0 < T$, run the randomized comparison design above for L_0 queries and then play $(\widehat{\theta}, \widehat{\theta})$. Bernstein's inequality with the larger budget, followed by a union bound, gives $\mathbb{P}(\widehat{\theta} \ne \theta) \le T^{-2}$; here $L_0 < T$ implies $k < T$, so $\log(eT)$ absorbs the union over coordinates. Exploration costs at most $2bL_0$, exploitation costs at most $2bT$ on the failure event and zero otherwise. Therefore

$$\mathfrak{R}^{\mathrm{pref}}_{k,H,\varepsilon,\beta}(T) \le C_F\,b\min\{T, (k + \alpha^{-2})\log(eT)\},$$

as claimed. ☐

D The fresh-direction lower bound

Proof of Theorem 6.2. Put

$$q(x) = \varepsilon + \frac{1}{2}\|x\|^2.$$

For $B \in S_2^{d-1}(1)$, $r \in [0,1]$, and $u = \langle \theta, B\rangle$,

$$|F_\theta(rB) - q(rB)| = \frac{1}{2}\left(s - \sqrt{1 + r^2 - 2ru}\right)_+^2.$$

Minimizing the square root over $r \in [0,1]$ shows that the supremum of the right side is $(s-1)^2/2$ when $u \leq 0$, and

$$\frac{1}{2}\left(s - \sqrt{1 - u^2}\right)^2$$

when $u > 0$. On $|u| \leq 1/2$, its square is at most $C(\varepsilon^4 + u^8)$; the complementary spherical cap has exponentially small probability. Rotation invariance and

$$\mathbb{E}_B\langle \theta, B\rangle^8 = \frac{105}{d(d+2)(d+4)(d+6)}$$

therefore give, uniformly in θ,

$$\mathbb{E}_B \sup_{0 \leq r \leq 1} |F_\theta(rB) - q(rB)|^2 \leq C(\varepsilon^4 + d^{-4}). \tag{74}$$

Now condition on an arbitrary posterior ν. The fresh Haar direction B is independent of Θ, and $R(B)$ is known after conditioning on B. Comparing the Gaussian observation channel with the Θ-independent reference $q(R(B)B) + \xi$, the information-radius bound gives

$$I_\nu(\Theta; B, F_\Theta(RB) + \xi) \leq \frac{1}{2}\mathbb{E}_{\Theta\sim\nu,B}|F_\Theta(RB) - q(RB)|^2.$$

Equation (74) proves (46).

Under the spherical prior, for a fixed nonadaptive design point $x = ru$, rotation invariance interchanges a fixed direction u under uniform Θ with a uniform direction B under fixed θ. Thus (74) also gives

$$\mathbb{E}_\Theta|F_\Theta(x) - q(x)|^2 \leq C(\varepsilon^4 + d^{-4}).$$

Condition on a possibly randomized design and compare its joint Gaussian observation law with the product reference $(q(X_i) + \xi_i)_{i \leq n}$. Additivity of the reference KL divergence proves (47).

For a stopped fresh-direction experiment, let $N_n = N \wedge n$ and use the cemetery-padded transcript. Since $\{N \geq t\}$ is H_{t-1}-measurable, conditional application of (46) and the chain rule give

$$I(\Theta; H_{N_n}) \leq C(\varepsilon^4 + d^{-4})\sum_{t=1}^{n}\mathbb{P}(N \geq t) = C(\varepsilon^4 + d^{-4})\,\mathbb{E}N_n.$$

Continuity of relative entropy along the increasing stopped σ-fields, followed by $n \to \infty$, proves (48).

For the converse, convexity at the origin and $\nabla F_\theta(0) = -b_0\theta$ give

$$F_\theta(x) \geq b_0\{1 - \langle \theta, x\rangle\}. \tag{75}$$

With $m_N = \mathbb{E}[\Theta \mid H_N]$, taking the infimum over x in (75) gives directly

$$r(Q_{H_N}) \geq b_0(1 - \|m_N\|).$$

Hence $r(Q_{H_N}) \leq \gamma b_0$ on M implies $\|m_N\| \geq 1-\gamma$. The uniform law σ on $S_2^{d-1}(1)$ is $1/(d-1)$-sub-Gaussian. The variational formula for relative entropy therefore yields

$$D_{\mathrm{KL}}(\nu\|\sigma) \geq \frac{d-1}{2}\|\mathbb{E}_\nu\Theta\|^2.$$

Apply this to each posterior and average on M:

$$I(\Theta; H_N) = \mathbb{E}D_{\mathrm{KL}}(Q_{H_N}\|\sigma) \geq \frac{d-1}{2}(1-\gamma)^2\mathbb{P}(\mathsf{M}).$$

Combining this inequality with (48) proves (50). □

Proof of the lower bound in Corollary 6.4. Let $N = \tau_\gamma \wedge T$ and $p = \mathbb{P}(\tau_\gamma \leq T)$. Before the contraction time, every action has conditional expected regret at least $r(Q_{t-1}) > \gamma b_0$. Hence

$$\mathbb{E}R_T \geq \gamma b_0\, \mathbb{E}N. \tag{76}$$

Moreover,

$$\mathbb{E}N \geq T(1-p).$$

Apply Theorem 6.2 to the stopped transcript H_N. On $\{\tau_\gamma \leq T\}$, its terminal posterior risk is at most γb_0, and therefore

$$Cd^{-4}\mathbb{E}N \geq I(\Theta; H_N) \geq c_\gamma d\, p$$

when $\varepsilon = d^{-1}$. Thus

$$\mathbb{E}N \geq c_\gamma d^5 p.$$

Minimizing $\max\{T(1-p), c_\gamma d^5 p\}$ over $p \in [0,1]$ gives $\mathbb{E}N \geq c_\gamma \min\{T, d^5\}$. Since $b_0 \asymp d^{-1}$, (76) proves (57). □

E The dimple regret bounds

Proof of the upper statements in Corollary 6.4. The same adaptive experiment, run to accuracy a and followed by commitment, gives the family upper bound. At $a = b_0 \asymp \varepsilon$, Theorem 6.3 uses $\widetilde{O}(d^2/\varepsilon^2)$ queries and $\widetilde{O}(d^2/\varepsilon)$ regret. Equation (55) proves the adaptive part of the query separation. Since $\varepsilon^4 + d^{-4} \leq 2\varepsilon^4$ in the stated range, Theorem 6.2 proves the lower part for nonadaptive experiments under the spherical prior and for fresh-direction experiments. Specializing to $\varepsilon = d^{-1}$ gives the displayed endpoint.

For the regret upper bound, first note that playing the origin for all T rounds costs $b_0 T \leq CT/d$. When $T \geq C_\gamma d^4 L_{d,T}^2$, choose

$$a = A_\gamma \frac{dL_{d,T}}{\sqrt{T}}$$

with A_γ large enough. Then $a \leq b_0$, $L_{d,a} \leq CL_{d,T}$, and (53) uses at most $T/2$ rounds. Indeed, enlarging the crossover constant makes $a \leq c/d \leq b_0$, while $1/a \leq \sqrt{T}/d$ and $\log(d/a) \leq \log(dT)$ give the displayed logarithmic comparison directly from (51). The exploration regret is at most

$$\frac{C_\gamma d^2 L_{d,a}^2}{a} \leq C_\gamma d L_{d,T}\sqrt{T}.$$

Play $\widehat{x}$ for the remaining rounds. Since losses are bounded by one, (52) gives

$$\sup_{\theta} \mathbb{E}_{\theta} F_{\theta}(\widehat{x}) \le (\gamma_0 + c_0)a \le C_{\gamma} a.$$

The exploitation regret is therefore at most $C_{\gamma} aT \le C_{\gamma} d L_{d,T} \sqrt{T}$. Below the displayed crossover, use the origin strategy. If $T \le d^4$, its cost CT/d is the smaller term in (58). If $d^4 < T < C_{\gamma} d^4 L_{d,T}^2$, then

$$\frac{T}{d} \le C_{\gamma} L_{d,T}\, d\sqrt{T}.$$

These two cases prove (58) throughout the remaining range. □

F Proof of the observation-adaptive experiment

Throughout this appendix, reduce ε_0, if necessary, so that $\varepsilon_0 \le 1/8$. Put

$$s_+ = \sqrt{1 + 2\varepsilon_0}, \qquad \beta_- = \frac{2}{s_+ + 1}.$$

Then, uniformly over the parameter range,

$$1 \le s \le s_+, \qquad \beta_- \varepsilon \le b_0 = s - 1 \le \varepsilon. \tag{77}$$

Fix

$$\gamma_0 = \frac{\gamma}{4}, \qquad c_{\delta} = \frac{\gamma}{4}, \qquad c_0 = c_{\delta}, \qquad \delta = c_{\delta} a,$$

so that $\gamma_0 + c_0 = \gamma/2 < \gamma$, and put

$$\mathcal{L} = 1 + \log\{16d(1 + \log(d/a))/\delta\}.$$

Thus $\mathcal{L} \le C_{\gamma} L_{d,a}$ after increasing $C_L(\gamma)$.

We now record the order in which the remaining constants are chosen. First choose a constant A_{H} large enough for all the Haar-coordinate bounds below. Next fix $C_{\mathrm{geo}} \ge 4$, and then choose

$$0 < q_{\star} \le \min\left\{\frac{1}{8}, \frac{1}{4C_{\mathrm{geo}}}\right\}. \tag{78}$$

In particular,

$$q_{\star} < s/4, \qquad C_{\mathrm{geo}} q_{\star} \le s/2$$

uniformly in ε. The initialization step c_h and its repetition constant are then chosen; this produces the numerical correlation constant $\kappa_{\mathrm{init}} > 0$ in (87) below. After κ_{init} is fixed, choose c_2; this fixes the geometric constants B_{λ}, c_4, k_0. Next choose k and $\rho = \sqrt{1 + k^2}$, and only then choose the amplification repetition constants C_{gate} and C_{amp}, whose simultaneous confidence bounds depend on the resulting deterministic iteration cap. Finally choose c_5, then c_{term}, then c_6, and last C_{con}. The compatible inequalities imposed on these constants are displayed at the point where each is used.

We use the following conditional form of sphere concentration. If $(u_i)_{i=1}^k$ is a Haar orthonormal basis of a fixed k-dimensional subspace and w is a fixed unit vector in that subspace, then, after increasing A_{H},

$$\max_{1 \le i \le k} |\langle w, u_i \rangle| \le A_{\mathrm{H}} \sqrt{\mathcal{L}/k} \tag{79}$$

except on an event of probability at most $\exp(-4\mathcal{L})$, provided $\mathcal{L} \le k$. When $\mathcal{L} > k$, we instead use the deterministic bound 1.

Every basis is drawn conditionally on the preceding transcript. Under $\mathbb{P}_\theta$, the current iterate and the tangent component of the fixed vector θ are then fixed, so (79) applies conditionally after normalizing a nonzero tangent component. Every empirical mean uses fresh Gaussian noise. The initialization, amplification, and contraction stages contain at most $C(1+\log(d/a))$ concentration calls in total. The definition of $\mathcal{L}$, together with sufficiently large repetition constants, therefore gives events $\mathcal{H}_{\text{init}}, \mathcal{H}_{\text{amp}}, \mathcal{H}_{\text{con}}$ such that, uniformly in θ,

$$\mathbb{P}_\theta(\mathcal{H}_{\text{init}}^c) \le \delta/3, \quad \mathbb{P}_\theta(\mathcal{H}_{\text{amp}}^c \mid \mathcal{H}_{\text{init}}) \le \delta/3, \quad \mathbb{P}_\theta(\mathcal{H}_{\text{con}}^c \mid \mathcal{H}_{\text{init}} \cap \mathcal{H}_{\text{amp}}) \le \delta/3.$$

Write $\mathcal{H} = \mathcal{H}_{\text{init}} \cap \mathcal{H}_{\text{amp}} \cap \mathcal{H}_{\text{con}}$. All three stages have deterministic caps. Off $\mathcal{H}$, the same capped routine is continued, returning the origin whenever a normalization is undefined and taking the stopping branch at every gate equality. These conventions make the complete experiment total and Borel measurable.

Whenever $r = \|\theta - x\| < s$, expanding (43) gives

$$F_\theta(x) = s\|\theta - x\| + \langle \theta, x \rangle - 1. \tag{80}$$

F.1 Orthogonal initialization

Lemma F.1. *With probability at least $1 - \delta/3$, the construction below returns a unit vector v_0 satisfying*

$$\langle \theta, v_0 \rangle \ge c \min\left\{1, \max\left(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d}\right)\right\}. \tag{81}$$

It uses at most $Cd^2\mathcal{L}^2/\varepsilon^2$ queries and incurs at most $Cd^2\mathcal{L}^2/\varepsilon$ regret on the success event.

Proof. Draw a Haar basis $(u_i)_{i=1}^d$, and set

$$h_0 = c_h \min\{\sqrt{\varepsilon}, \varepsilon\sqrt{d/\mathcal{L}}\}, \qquad n_0 = \left\lceil C_{\text{init}} d\mathcal{L}^2/\varepsilon^2 \right\rceil. \tag{82}$$

Estimate every centered difference

$$D_i^0 = \frac{F_\theta(h_0 u_i) - F_\theta(-h_0 u_i)}{2h_0}$$

using n_0 repetitions at each of the two points. Let A_0 be a universal constant for the Taylor estimate used below. If $\mathcal{L} \le d$, then $h_0 = c_h\varepsilon/t$, where

$$t = \max\{\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d}\},$$

and (79) gives

$$h_0^2 + 2h_0 \max_i |\langle \theta, u_i \rangle| \le (c_h^2 + 2A_{\text{H}} c_h)\varepsilon.$$

If $\mathcal{L} > d$, then $h_0 = c_h\varepsilon\sqrt{d/\mathcal{L}}$, and the deterministic coordinate bound gives the same inequality with A_{H} replaced by one. Choose $c_h \le 1/8$ so that

$$c_h^2 + 2\max\{A_{\text{H}}, 1\}c_h \le 1, \qquad A_0 c_h^2 \le \frac{\beta_-}{8}. \tag{83}$$

Both query points are then strictly in the active dimple. Equation (80) gives

$$D_i^0 = -\lambda_i \langle \theta, u_i \rangle, \qquad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1, \qquad |\lambda_i - b_0| \le A_0 h_0^2, \tag{84}$$

where $r_i^{\pm} = \|\theta \mp h_0 u_i\|$. Hence the noiseless difference vector is

$$D = -b_0\theta + e_0, \qquad \|e_0\| \le A_0 h_0^2 \le \frac{\beta_-}{8}\varepsilon.$$

Write $\widehat{D} = D + \zeta$. Conditional on the basis, ζ is an isotropic Gaussian vector with coordinate variance $(2n_0h_0^2)^{-1}$. On $\mathcal{H}_{\text{init}}$,

$$|\langle\theta, \zeta\rangle| \le \frac{C\varepsilon}{\sqrt{C_{\text{init}}}\, h_0\sqrt{d\mathcal{L}}}, \tag{85}$$

$$\|\zeta\| \le \frac{C\varepsilon}{\sqrt{C_{\text{init}}}\, h_0\mathcal{L}} \left(1 + \sqrt{\mathcal{L}/d}\right). \tag{86}$$

Set

$$t = \min\{1, \max(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d})\}.$$

If $\mathcal{L} \le d$, then $h_0 = c_h\varepsilon/t$, and

$$\frac{t}{\sqrt{d\mathcal{L}}} \le \varepsilon, \qquad \frac{t^2}{\mathcal{L}}\left(1 + \sqrt{\mathcal{L}/d}\right) \le 2\varepsilon.$$

These inequalities follow by considering separately $t^2 = \varepsilon$ and $t^2 = \mathcal{L}/d$. If $\mathcal{L} > d$, then $t = 1$, $h_0 = c_h\varepsilon\sqrt{d/\mathcal{L}}$, and

$$\frac{1}{d} \le \varepsilon, \qquad \frac{1 + \sqrt{\mathcal{L}/d}}{\sqrt{d\mathcal{L}}} \le \frac{2}{d} \le 2\varepsilon.$$

Consequently, after choosing C_{init} sufficiently large,

$$|\langle\theta, \zeta\rangle| \le \frac{\beta_-}{8}\varepsilon, \qquad \|\zeta\| \le B_{\text{init}}\frac{\varepsilon}{t}$$

for a fixed universal B_{init}. Together with (77) and (83), this yields

$$\langle\theta, -\widehat{D}\rangle \ge \frac{\beta_-}{2}\varepsilon, \qquad \|\widehat{D}\| \le (2 + B_{\text{init}})\,\frac{\varepsilon}{t}.$$

Thus $\widehat{D} \ne 0$, and, with

$$\kappa_{\text{init}} = \frac{\beta_-}{2(2 + B_{\text{init}})}, \qquad c_{\text{init}} := \kappa_{\text{init}} t,$$

the vector $v_0 = -\widehat{D}/\|\widehat{D}\|$ satisfies

$$\langle\theta, v_0\rangle \ge c_{\text{init}} = \kappa_{\text{init}}\min\{1, \max(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d})\}. \tag{87}$$

This implies (81).

The stage uses $2dn_0 = O(d^2\mathcal{L}^2/\varepsilon^2)$ queries. Every query point has loss $O(\varepsilon)$, so its regret is $O(d^2\mathcal{L}^2/\varepsilon)$. $\square$

F.2 Tangent amplification

Lemma F.2. *Conditionally on the initialization event, a capped tangent routine reaches* $\|\theta - v\| \le q_\star$ *with failure probability at most* $\delta/3$. *With* c_{init} *defined in* (87), *the stage uses*

$$N_{\text{amp}} \le \frac{Cd^2\mathcal{L}}{c_{\text{init}}^4}, \qquad R_{\text{amp}} \le \frac{Cd^2\mathcal{L}}{c_{\text{init}}^2}, \tag{88}$$

up to lower-order $C\mathcal{L}(1+\log(1/c_{\text{init}}))$ *gate costs.*

Proof. Let

$$\phi_\varepsilon(q) = \varepsilon + \frac{1}{2} - \frac{1}{2}(s-q)_+^2, \qquad \Delta_\star = \phi_\varepsilon(q_\star) - \phi_\varepsilon(q_\star/2).$$

Because $q_\star < s$,

$$\Delta_\star = \frac{sq_\star}{2} - \frac{3q_\star^2}{8} \ge \frac{q_\star}{2} - \frac{3q_\star^2}{8} > 0. \tag{89}$$

The lower bound is universal.

The routine maintains a unit vector v_j and a deterministic number $\underline{c}_j$ satisfying

$$\langle \theta, v_j \rangle \ge \underline{c}_j, \qquad \underline{c}_j = \rho^j c_{\text{init}}, \tag{90}$$

where $\rho > 1$ is fixed below. If $\underline{c}_j > 1 - q_\star^2/8$, then

$$\|\theta - v_j\|^2 = 2\{1 - \langle \theta, v_j \rangle\} < q_\star^2/4,$$

and the routine stops.

Otherwise, average $m_{\text{gate}} = \lceil C_{\text{gate}}\mathcal{L} \rceil$ observations at v_j, and compare the average with

$$T_{\text{gate}} = \frac{\phi_\varepsilon(q_\star/2) + \phi_\varepsilon(q_\star)}{2}.$$

The routine stops when the average is at most T_{gate}, and continues otherwise. After k and ρ are fixed below, choose C_{gate} large enough so that, simultaneously at every gate,

$$|\overline{Y}_j - F_\theta(v_j)| \le \Delta_\star/4$$

on $\mathcal{H}_{\text{amp}}$. Since $F_\theta(v_j) = \phi_\varepsilon(\|\theta - v_j\|)$ and ϕ_ε is nondecreasing, stopping implies

$$\|\theta - v_j\| \le q_\star, \tag{91}$$

whereas continuing implies

$$\|\theta - v_j\| \ge q_\star/2. \tag{92}$$

Suppose the gate continues. Write

$$c = \underline{c}_j, \qquad v = v_j, \qquad p = \langle \theta, v \rangle, \qquad x = (c/2)v, \qquad r_0 = \|\theta - x\|.$$

Then $p \ge c$, and

$$r_0^2 = 1 + \frac{c^2}{4} - cp \le 1 - \frac{3c^2}{4}, \qquad r_0 \ge 1 - \frac{c}{2} \ge \frac{1}{2}. \tag{93}$$

$$c^2 \ge c_{\text{init}}^2 \ge \kappa_{\text{init}}^2 \varepsilon. \tag{94}$$

Draw a Haar orthonormal basis $(u_i)_{i=1}^{d-1}$ of $v^{\perp}$, and write $q_i = \langle \theta, u_i \rangle$. When $\mathcal{L} \le d-1$, set $h = c_2 c$. On the conditional Haar event,

$$|q_i| \le A_{\mathrm{H}} \sqrt{\mathcal{L}/(d-1)} \le \frac{2A_{\mathrm{H}}}{\kappa_{\mathrm{init}}} c.$$

When $\mathcal{L} > d-1$, set $h = c_2 c^2$ and use $|q_i| \le 1$; in this case $\mathcal{L}/d \ge 1/2$, so $c \ge c_{\mathrm{init}} \ge \kappa_{\mathrm{init}}/\sqrt{2}$. Choose $c_2 \le 1/8$ so that

$$c_2^2 + \frac{4A_{\mathrm{H}}}{\kappa_{\mathrm{init}}} c_2 \le \frac{3}{8}, \qquad c_2^2 + 2c_2 \le \frac{3}{8}. \tag{95}$$

In the two cases, respectively,

$$h^2 + 2h|q_i| \le \frac{3c^2}{8}.$$

Together with (93), this shows that $x \pm h u_i$ lie in the active dimple. Their norms are at most one after decreasing c_2, if necessary.

For $r_i^{\pm} = \|\theta - x \mp h u_i\|$, the exact centered difference is

$$D_i = \frac{F_\theta(x + hu_i) - F_\theta(x - hu_i)}{2h} = -\lambda_i q_i, \qquad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1. \tag{96}$$

Let $\lambda_0 = s/r_0 - 1$. From (93), for a universal constant $B_0 < \infty$,

$$\lambda_0 \ge 1 - r_0 \ge \frac{3c^2}{8}, \qquad \lambda_0 \le B_0 c, \tag{97}$$

where the upper bound uses $s - 1 \le \varepsilon \le c^2/\kappa_{\mathrm{init}}^2$ and $r_0 \ge 1/2$. A Taylor expansion of $\sqrt{r_0^2 + z}$, uniform over the preceding activity margin, gives

$$|\lambda_i - \lambda_0| \le A_\lambda h^2$$

for a universal A_λ. Decrease c_2 further so that

$$A_\lambda c_2^2 \le \frac{3}{16}. \tag{98}$$

Since $c \le 1$, both choices of h then imply

$$\frac{3c^2}{16} \le \lambda_i \le B_\lambda c \tag{99}$$

for a universal B_λ.

Set

$$g = -\sum_{i=1}^{d-1} D_i u_i = \sum_{i=1}^{d-1} \lambda_i q_i u_i.$$

If $q = \|\theta - v\|$, then $p = 1 - q^2/2$, and (92) gives

$$\sum_i q_i^2 = 1 - p^2 = q^2(1 - q^2/4) \ge q_\star^2/8.$$

Therefore (99) implies

$$\langle \theta, g \rangle \ge c_4 c^2, \qquad \|g\| \le B_\lambda c, \qquad c_4 := \frac{3q_\star^2}{128}. \tag{100}$$

Estimate every D_i with

$$n_c = \left\lceil C_{\mathrm{amp}} d\mathcal{L}/c^4 \right\rceil$$

repetitions at each of $x + hu_i$ and $x - hu_i$. With empirical differences $\widehat{D}_i$, define

$$\widehat{g} = -\sum_{i=1}^{d-1} \widehat{D}_i u_i = g + \zeta.$$

Conditional on the basis, ζ is isotropic in $v^{\perp}$, with coordinate variance $(2n_c h^2)^{-1}$. For $\mathcal{L} \le d-1$, this variance is at most

$$\frac{Cc^2}{C_{\mathrm{amp}} d\mathcal{L}};$$

for $\mathcal{L} > d-1$, it is at most

$$\frac{C}{C_{\mathrm{amp}} d\mathcal{L}},$$

and c is bounded below by a universal multiple of κ_{init}. After k and ρ are fixed below, choose C_{amp} large enough so that, simultaneously at all stages,

$$\|\zeta\| \le c, \qquad |\langle \theta, \zeta \rangle| \le \frac{c_4}{2} c^2 \tag{101}$$

on $\mathcal{H}_{\mathrm{amp}}$.

It follows that $\widehat{g} \ne 0$, and for $u = \widehat{g}/\|\widehat{g}\|$,

$$\langle \theta, u \rangle \ge k_0 c, \qquad k_0 := \frac{c_4}{2(B_\lambda + 1)}.$$

Choose

$$0 < k \le \min\{k_0/2, q_\star/4\}, \qquad \rho = \sqrt{1+k^2} > 1, \tag{102}$$

and set

$$v_{j+1} = \frac{v_j + ku}{\sqrt{1+k^2}}, \qquad \underline{c}_{j+1} = \rho \underline{c}_j.$$

Because $u \perp v_j$,

$$\langle \theta, v_{j+1} \rangle \ge \frac{c + kk_0 c}{\sqrt{1+k^2}} \ge \sqrt{1+k^2}\, c = \underline{c}_{j+1}.$$

Thus (90) is preserved. The restriction $k \le q_\star/4$ also ensures $\rho(1 - q_\star^2/8) < 1$, so the deterministic lower bounds never exceed one before the next stopping check.

After at most

$$1 + \left\lceil \frac{\log\{(1 - q_\star^2/8)/c_{\mathrm{init}}\}}{\log \rho} \right\rceil = O(1 + \log(1/c_{\mathrm{init}}))$$

iterations, either a gate has stopped or the deterministic correlation threshold has been crossed. In both cases the returned vector satisfies $\|\theta - v\| \le q_\star$.

At correlation level c, the tangent queries number $O(d^2\mathcal{L}/c^4)$. By (94),

$$F_\theta(x \pm hu_i) \le \varepsilon + \frac{1}{2}\|x \pm hu_i\|^2 \le Cc^2,$$

so their regret is $O(d^2\mathcal{L}/c^2)$. The gate uses $O(\mathcal{L})$ queries and regret per iteration. Geometric summation over $c_j = \rho^j c_{\mathrm{init}}$ gives

$$N_{\mathrm{amp}} \le \frac{Cd^2\mathcal{L}}{c_{\mathrm{init}}^4} + C\mathcal{L}(1 + \log(1/c_{\mathrm{init}})),$$

and

$$R_{\text{amp}} \leq \frac{Cd^2\mathcal{L}}{c_{\text{init}}^2} + C\mathcal{L}(1 + \log(1/c_{\text{init}})).$$

Finally,

$$c_{\text{init}} \geq \kappa_{\text{init}} \max\{\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d}\}$$

unless it is already a fixed constant. This yields (88) and the simplified bounds

$$O(d^2\mathcal{L}/\varepsilon^2) \quad \text{and} \quad O(d^2\mathcal{L}/\varepsilon),$$

respectively. □

F.3 Angular contraction

Lemma F.3. *Conditionally on the success events of the first two stages, the capped contraction stage returns $\widehat{x}$ satisfying $F_\theta(\widehat{x}) \leq \gamma_0 a$ with conditional probability at least $1 - \delta/3$. It has deterministic query cap*

$$C_\gamma d^2\mathcal{L}/a^2$$

and success-event regret at most

$$C_\gamma d^2\mathcal{L}/a.$$

Proof. Choose

$$0 < c_5 \leq \frac{1}{8}, \qquad 0 < c_{\text{term}} \leq \frac{\gamma_0}{s_+}. \tag{103}$$

Let

$$Q_j = q_\star 2^{-j}, \qquad J = \min\{j \geq 0 : Q_j \leq c_{\text{term}} a\}.$$

The routine maintains the invariant

$$\|v_j\| = 1, \qquad \|\theta - v_j\| \leq Q_j. \tag{104}$$

The amplification lemma establishes it at $j = 0$. For each $j < J$, write $v = v_j$, $Q = Q_j$, set $h = c_5 Q$, and take a measurably chosen orthonormal basis $(u_i)_{i=1}^{d-1}$ of $v^\perp$. Define

$$\alpha_h = (1 + h^2)^{-1/2}, \qquad t = h(1 + h^2)^{-1/2}, \qquad x_i^\pm = \alpha_h v \pm t u_i.$$

Then $x_i^\pm$ are unit vectors, $t \geq c_5 Q/\sqrt{2}$, and

$$\|x_i^\pm - v\|^2 = 2(1 - \alpha_h) \leq h^2.$$

Consequently,

$$\|\theta - x_i^\pm\| \leq (1 + c_5)Q \leq C_{\text{geo}} Q \leq C_{\text{geo}} q_\star \leq s/2. \tag{105}$$

Thus every probe lies in the active dimple.

Write

$$\theta = c_v v + w, \qquad c_v = \langle \theta, v \rangle = 1 - \frac{1}{2}\|\theta - v\|^2, \qquad w = \sum_i q_i u_i.$$

Since $Q \leq q_\star \leq 1/8$, one has $c_v > 0$,

$$c_v = \sqrt{1 - \|w\|^2}, \qquad \|w\| \leq \|\theta - v\| \leq Q < 1/2. \tag{106}$$

Let

$$\psi(r) = sr - \frac{1}{2}r^2, \qquad G(p) = \psi(\sqrt{2-2p}).$$

Define

$$I_Q = \{p \in [-1,1] : \sqrt{2-2p} \le C_{\mathrm{geo}} Q\}.$$

By (105), the inner products

$$p_i^\pm = \langle \theta, x_i^\pm \rangle = \alpha_h c_v \pm t q_i$$

belong to I_Q, and $F_\theta(x_i^\pm) = G(p_i^\pm)$. The restriction $G_Q := G|_{I_Q}$ is strictly decreasing. Moreover, for $p \in I_Q$,

$$|(G_Q^{-1})'(G(p))| = \frac{\sqrt{2-2p}}{s - \sqrt{2-2p}} \le 2C_{\mathrm{geo}} Q. \tag{107}$$

The right side tends to zero at $p = 1$, so this estimate includes the endpoint by continuity. Set

$$B_q := \frac{2\sqrt{2}C_{\mathrm{geo}}}{c_5},$$

and choose $c_6 > 0$ so that

$$2B_q c_6 \le \frac{1}{2}. \tag{108}$$

Let $\overline{Y}_i^\pm$ be the sample means at $x_i^\pm$, and let $\widetilde{Y}_i^\pm$ be their Euclidean projections onto the interval $G(I_Q)$. Define

$$\widehat{q}_i = \frac{G_Q^{-1}(\widetilde{Y}_i^+) - G_Q^{-1}(\widetilde{Y}_i^-)}{2t}, \qquad \widehat{w} = \sum_i \widehat{q}_i u_i.$$

Use

$$n_Q = \lceil C_{\mathrm{con}} d\mathcal{L}/Q^2 \rceil$$

repetitions at each site. Choose C_{con} so that, simultaneously for all i and all $j < J$,

$$|\overline{Y}_i^\pm - F_\theta(x_i^\pm)| \le \frac{c_6 Q}{\sqrt{d}} \tag{109}$$

on $\mathcal{H}_{\mathrm{con}}$. Projection onto $G(I_Q)$ cannot increase the error because the true value belongs to this interval. Combining (107), (109), and $t \ge c_5 Q/\sqrt{2}$ gives

$$|\widehat{q}_i - q_i| \le B_q \frac{c_6 Q}{\sqrt{d}},$$

and hence

$$\|\widehat{w} - w\| \le B_q c_6 Q. \tag{110}$$

Let $\widetilde{w}$ be the Euclidean projection of $\widehat{w}$ onto the ball of radius $1/2$. Since the true w belongs to this ball by (106),

$$\|\widetilde{w} - w\| \le \|\widehat{w} - w\|.$$

Set

$$v_{j+1} = \sqrt{1 - \|\widetilde{w}\|^2}\, v + \widetilde{w}.$$

The map

$$z \longmapsto \sqrt{1 - \|z\|^2}\, v + z$$

is 2-Lipschitz on the ball of radius 1/2. Using $\theta = \sqrt{1 - \|w\|^2}\, v + w$, we obtain

$$\|\theta - v_{j+1}\| \le 2\|\widetilde{w} - w\| \le Q/2 = Q_{j+1}.$$

This proves (104) by induction.

The number of queries at scale Q_j is $O(d^2\mathcal{L}/Q_j^2)$. By (105),

$$F_\theta(x_i^\pm) = \psi(\|\theta - x_i^\pm\|) \le s_+ C_{\mathrm{geo}} Q_j,$$

so the corresponding regret is $O(d^2\mathcal{L}/Q_j)$. If $J = 0$, this stage makes no queries. Otherwise, Q_j decreases geometrically and $Q_{J-1} > c_{\mathrm{term}} a$, so

$$\sum_{j<J} \frac{d^2\mathcal{L}}{Q_j^2} \le \frac{Cd^2\mathcal{L}}{c_{\mathrm{term}}^2 a^2}, \qquad \sum_{j<J} \frac{d^2\mathcal{L}}{Q_j} \le \frac{Cd^2\mathcal{L}}{c_{\mathrm{term}} a}.$$

The output $\widehat{x} = v_J$ then satisfies

$$F_\theta(\widehat{x}) = \psi(\|\theta - v_J\|) \le s_+\|\theta - v_J\| \le s_+ c_{\mathrm{term}} a \le \gamma_0 a.$$

The number J and every repetition count are deterministic. Continuing the same clipped updates off $\mathcal{H}_{\mathrm{con}}$ therefore gives the claimed deterministic cap. □

Proof of Theorem 6.3. Run Lemmas F.1–F.3 in order, assigning failure probability $\delta/3$ to each. The three conditional failure bounds imply $\sup_\theta \mathbb{P}_\theta(\mathcal{H}^c) \le \delta = c_0 a$. The amplification lower bounds and the angular invariant are asserted only on $\mathcal{H}$; the deterministic caps, clipping rules, and fixed iteration limits define the same routine on $\mathcal{H}^c$. Because $a \le b_0 \asymp \varepsilon$, their deterministic caps and geometric sums give

$$N \le C_\gamma d^2 L_{d,a}^2/a^2$$

and, on $\mathcal{H}$,

$$R_N \le C_\gamma d^2 L_{d,a}^2/a.$$

On $\mathcal{H}^c$, the capped routine uses no more than its displayed query budget and incurs at most one unit of regret per query. Since $\mathbb{P}(\mathcal{H}^c) \le \delta = c_\delta a$, this adds at most

$$C_\gamma d^2 L_{d,a}^2/a$$

to expected regret. The accuracy statement follows from Lemma F.3 and the union bound. Since losses are bounded by one, the same statement gives, under every prior,

$$\mathbb{E}F_\Theta(\widehat{x}) \le (\gamma_0 + c_0)a.$$

Apply Markov's inequality to the nonnegative random variable $\mathbb{E}[F_\Theta(\widehat{x}) \mid H_N]$ to obtain (55). On the displayed event, the posterior Bayes risk is no larger than the posterior loss of $\widehat{x}$, and hence is at most γa. □

G Gaussian models and Bayes–minimax equality

This appendix proves Theorem 5.1. It establishes measurable selection of statewise experiments and finite-horizon Bayes–minimax equality; Section 2 already proved the two-sided stopped-cost conversion.

Proof of Corollary 5.3. The upper endpoint is the current general BCO upper bound (Lattimore and György, 2023; Fokkema et al., 2024); the lower endpoint is the minimax theorem of Rajaraman (2026). Its 1-Lipschitz hard family is on the Euclidean unit ball and is anchored by $f(0) = 0$. Consequently $f(x) \in [-1, 1]$, so it embeds in the present bounded unit-noise model by the affine rescaling $g = (f + 1)/2 \in [0, 1]$. From an observation $f(x) + \xi$, return $(f(x) + \xi + 1)/2 + \zeta$, where $\zeta \sim N(0, 3/4)$ is independent. The result is $g(x) + \xi'$ with $\xi' \sim N(0, 1)$, and regret changes only by a factor two. Apply Corollary 5.2. $\square$

Lemma G.1 (Prior-wise Borel realization)**.** *Fix a Borel prior Q and a finite horizon T. On standard Borel history, action, and observation spaces, every universally measurable randomized nonanticipating policy has a Borel randomized nonanticipating policy inducing the same joint law of (F, H_T) under Q. The same holds when the action space is augmented by a stopping symbol.*

Proof. At time t, represent the action kernel as a universally measurable map from histories into the standard Borel space of action distributions. Under the reached history law, this map has a Borel version. Replace it by that version; the strategic law through time t is unchanged. Induction over $t = 1, \dots, T$ proves the claim. Adding one isolated stopping symbol preserves the standard Borel property. $\square$

G.1 Finite-horizon Bayes–minimax identity

Let $\nu = N(0, 1)$, let $\mathsf{H}_T = (C \times \mathbb{R})^T$, and equip $\mathcal{P}(\mathsf{H}_T)$ with the topology of weak convergence. Let $\mathcal{P}_T$ be the set of reference strategic measures

$$P(dh_T) = \prod_{t=1}^{T} \pi_t(dx_t \mid h_{t-1})\, \nu(dy_t)$$

generated by Borel randomized nonanticipating policies.

Lemma G.2 (Compactness of reference strategic measures)**.** *Suppose that C is a nonempty compact metric space. Then $\mathcal{P}_T$ is a nonempty convex weakly compact subset of $\mathcal{P}(\mathsf{H}_T)$.*

More precisely, a probability measure P on H_T belongs to $\mathcal{P}_T$ if and only if, for every $t \le T$, $a \in C_b((C \times \mathbb{R})^{t-1} \times C)$, and $b \in C_b(\mathbb{R})$,

$$\int a(h_{t-1}, x_t) b(y_t)\, dP = \left(\int b\, d\nu \right) \int a(h_{t-1}, x_t)\, dP \tag{111}$$

Proof. Every reference strategic measure satisfies (111) by iterated conditioning. Conversely, the bounded-continuous identities extend, by a measure-determining and monotone-class argument on Polish spaces, to bounded Borel a and b. They therefore imply

$$P(Y_t \in \cdot \mid H_{t-1}, X_t) = \nu(\cdot) \qquad P\text{-a.s.}$$

Because all spaces are standard Borel, P has Borel regular conditional distributions

$$\pi_t(\cdot \mid h_{t-1}) = P(X_t \in \cdot \mid H_{t-1} = h_{t-1}).$$

Successive disintegration, together with the displayed conditional law of Y_t, gives

$$P(dh_T) = \prod_{t=1}^{T} \pi_t(dx_t \mid h_{t-1})\, \nu(dy_t),$$

so $P \in \mathcal{P}_T$. This proves the characterization.

The characterization is linear in P, so $\mathcal{P}_T$ is convex. It also implies that the $Y_{1:T}$-marginal of every $P \in \mathcal{P}_T$ is $\nu^{\otimes T}$. Since C^T is compact, this common observation marginal makes $\mathcal{P}_T$ tight.

If $P_n \in \mathcal{P}_T$ and $P_n \Rightarrow P$, then both sides of (111) converge for every bounded continuous a, b. Hence P satisfies the same identities and, by the characterization, belongs to $\mathcal{P}_T$. Thus $\mathcal{P}_T$ is weakly closed. Prokhorov's theorem now gives weak compactness. $\square$

For $f \in \mathcal{C}(C, [0,1])$, write

$$m(f) = \min_{x \in C} f(x), \qquad G_f(h_T) = \sum_{t=1}^{T} \{f(x_t) - m(f)\},$$

and

$$L_f(h_T) = \exp\left\{\sum_{t=1}^{T}\left(y_t f(x_t) - \frac{1}{2} f(x_t)^2\right)\right\},$$

and define

$$u(P, f) = \int L_f(h_T) G_f(h_T)\, P(dh_T).$$

Lemma G.3 (Continuity of the Gaussian regret payoff)**.** *The map*

$$(P, f) \longmapsto u(P, f)$$

is jointly continuous on $\mathcal{P}_T \times \mathcal{C}(C, [0,1])$, where the first space has the weak topology and the second has the uniform topology. It is affine in P and takes values in $[0, T]$.

If P is the reference strategic measure generated by a policy π, then

$$L_f = \frac{dP_f^\pi}{dP} \quad \text{on } \mathsf{H}_T, \qquad u(P, f) = \mathbb{E}_f^\pi R_T.$$

Proof. The likelihood-ratio identity follows by multiplying the one-step density ratios

$$\frac{\varphi(y_t - f(x_t))}{\varphi(y_t)} = \exp\left\{y_t f(x_t) - \frac{1}{2} f(x_t)^2\right\}.$$

The action kernels cancel because the same policy is used under the reference and shifted observation laws. Thus $L_f P$ is the actual strategic law, and

$$u(P, f) = \mathbb{E}_f^\pi R_T \in [0, T].$$

Fix f. The integrand $L_f G_f$ is continuous on H_T, but it is unbounded. Put

$$S(y_{1:T}) = \sum_{t=1}^{T} |y_t|.$$

Since $0 \le G_f \le T$,

$$0 \le L_f(h_T) G_f(h_T) \le T e^{S(y_{1:T})}. \tag{112}$$

The right-hand side is integrable under $\nu^{\otimes T}$, and every $P \in \mathcal{P}_T$ has this same observation marginal. Let $\chi_M \in C_c(\mathbb{R}^T)$ satisfy $0 \le \chi_M \le 1$ and $\chi_M(y) \to 1$ pointwise. Then $\chi_M(y_{1:T}) L_f(h_T) G_f(h_T)$ is bounded and continuous, while

$$\sup_{P \in \mathcal{P}_T} \int (1 - \chi_M) L_f G_f \, dP \le T \int (1 - \chi_M(y)) e^{S(y)}\, \nu^{\otimes T}(dy) \longrightarrow 0.$$

It follows that weak convergence $P_n \Rightarrow P$ implies $u(P_n, f) \to u(P, f)$.

It remains to control changes in f, uniformly over P. Let $\delta = \|f - f'\|_\infty$. Since

$$|m(f) - m(f')| \le \delta, \qquad |G_f - G_{f'}| \le 2T\delta,$$

and, upon writing $A_f = \log L_f$,

$$|A_f - A_{f'}| \le \delta\{S(y_{1:T}) + T\}, \qquad A_f, A_{f'} \le S(y_{1:T}),$$

the mean-value theorem gives

$$|L_f G_f - L_{f'} G_{f'}| \le T\delta e^{S(y_{1:T})}\{S(y_{1:T}) + T + 2\}.$$

Therefore

$$\sup_{P \in \mathcal{P}_T} |u(P, f) - u(P, f')| \le K_T \|f - f'\|_\infty,$$

where

$$K_T = T \int e^{S(y)}\{S(y) + T + 2\}\, \nu^{\otimes T}(dy) < \infty.$$

Combining this uniform estimate with continuity in P for fixed f proves joint continuity. Affinity in P is immediate. □

Proof of (39). Fix $C \in \mathcal{C}_d$ and abbreviate $\mathfrak{F} = \mathfrak{F}_d(C)$. Let $\mathcal{V}_{\mathrm{fs}}$ be the real vector space of finitely supported signed measures on $\mathfrak{F}$, equipped with the locally convex topology generated by the seminorms

$$q \longmapsto \left| \int \phi(f)\, q(df) \right|, \qquad \phi \in C_b(\mathfrak{F}).$$

Let $\mathcal{Q}_0 \subset \mathcal{V}_{\mathrm{fs}}$ be the convex set of finitely supported probability measures, and define

$$U(P, Q) = \int_{\mathfrak{F}} u(P, f)\, Q(df).$$

By Lemma G.3, $f \mapsto u(P, f)$ is bounded and continuous. Hence $Q \mapsto U(P, Q)$ is continuous and affine on $\mathcal{Q}_0$. For each $Q \in \mathcal{Q}_0$, $P \mapsto U(P, Q)$ is a finite convex combination of continuous affine functions and is therefore continuous and affine. Lemma G.2 makes $\mathcal{P}_T$ compact and convex. Sion's theorem (Sion, 1958) gives

$$\min_{P \in \mathcal{P}_T} \sup_{Q \in \mathcal{Q}_0} U(P, Q) = \sup_{Q \in \mathcal{Q}_0} \min_{P \in \mathcal{P}_T} U(P, Q).$$

Because every Dirac measure belongs to $\mathcal{Q}_0$,

$$\sup_{Q \in \mathcal{Q}_0} U(P, Q) = \sup_{f \in \mathfrak{F}} u(P, f).$$

The preceding lemmas identify every $P \in \mathcal{P}_T$ with a Borel randomized nonanticipating policy and identify $u(P, f)$ with that policy's regret under f. Consequently,

$$\inf_{\pi} \sup_{f \in \mathfrak{F}} \mathbb{E}_f^{\pi} R_T = \sup_{Q \in \mathcal{Q}_0} \inf_{\pi} \mathbb{E}_Q^{\pi} R_T. \tag{113}$$

For an arbitrary Borel prior Q on $\mathfrak{F}$, if $P_n \Rightarrow P$, Lemma G.3 gives $u(P_n, f) \to u(P, f)$ for every f. Since $0 \le u \le T$, dominated convergence makes $P \mapsto \int u(P, f)\, Q(df)$ continuous. Compactness of $\mathcal{P}_T$ therefore attains the minimum, and

$$\min_{P \in \mathcal{P}_T} \int u(P, f)\, Q(df) \le \min_{P \in \mathcal{P}_T} \sup_{f \in \mathfrak{F}} u(P, f).$$

Taking the supremum over all Borel priors cannot exceed the minimax value, whereas, by (113), the supremum over finitely supported priors already equals the minimax value. Hence

$$\inf_{\pi} \sup_{f \in \mathfrak{F}} \mathbb{E}_f^{\pi} R_T = \sup_{Q \text{ on } \mathfrak{F}} \inf_{\pi} \mathbb{E}_Q^{\pi} R_T,$$

and the supremum on the right may be restricted to finitely supported priors. Taking the outer supremum over $C \in \mathcal{C}_d$ proves (39). $\square$

G.2 Measurable selection of stopped experiments

Lemma G.4 (Capped Gaussian selection)**.** *Fix a query cap $L < \infty$. The capped stopped value*

$$v_L(Q, b) = \inf_{\mathcal{E}: N \le L} \frac{b\, \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})}$$

is universally measurable. For every $\epsilon > 0$, there is a universally measurable state-to-experiment rule with value at most $v_L(Q, b) + \epsilon$ at every active state.

Proof. The belief space $\mathcal{P}(\mathfrak{F}_d(C))$ is standard Borel. For a belief μ, define

$$\ell(\mu, x) = \int \{f(x) - \min_C f\}\, \mu(df), \qquad r(\mu) = \min_{x \in C} \ell(\mu, x).$$

The first map is Borel and continuous in x; compactness of C makes r Borel and supplies a Borel Bayes-action selector. With φ the standard Gaussian density, the posterior update is the Borel map

$$\Psi(\mu, x, o)(df) = \frac{\varphi(o - f(x))\, \mu(df)}{\int \varphi(o - f'(x))\, \mu(df')}.$$

For a rational price $q > 0$, the capped penalized problem is exactly the recursion (7), with $\mu' = \Psi(\mu, x, o)$. Its one-step costs are bounded below and all primitive kernels are Borel. Standard integration, partial minimization, and universally measurable selection for lower-semianalytic dynamic programs therefore apply (Bertsekas and Shreve, 1978, Propositions 7.47–7.50). In particular, its root value $J_L^q(\mu, b)$ is lower semianalytic and has universally measurable near-minimizers.

Directly from the definition,

$$v_L(\mu, b) < q \quad \Longleftrightarrow \quad J_L^q(\mu, b) < 0.$$

The rational strict sublevel sets make v_L universally measurable. On $\{v_L < +\infty\}$, enumerate the positive rationals and measurably choose the first q with

$$v_L + \epsilon/4 < q < v_L + \epsilon/2.$$

Then $J_L^q < 0$. Partition this set further by the first integer n such that $J_L^q \le -1/n$, and use a Bellman selector whose approximation error is less than $1/(2n)$. The resulting penalized value remains negative, so its success probability is positive and its ratio is below $q < v_L + \epsilon/2$. On $\{v_L = +\infty\}$, any measurable one-query rule satisfies the extended-value approximation. Countable patching gives one universally measurable ϵ-optimal rule. $\square$

Lemma G.5 (Removing the query cap)**.** *The unrestricted value* $\mathsf{SC}_b(Q)$ *is universally measurable and has a universally measurable* ϵ*-optimal selector for every* $\epsilon > 0$*. The selected experiment may have a finite deterministic cap* $L(Q, b)$ *depending on the state.*

Proof. Truncate an admissible experiment after L queries. As $L \to \infty$, its stopped regret and hit probability converge monotonically to those of the complete experiment:

$$\mathbb{E}R_{N\wedge L} \uparrow \mathbb{E}R_N, \qquad \mathbb{P}(\mathsf{M}_L) \uparrow \mathbb{P}(\mathsf{M}).$$

Hence its ratio converges, including the zero-success case under the $+\infty$ convention. Indeed, if $\mathbb{P}(\mathsf{M}) > 0$, componentwise convergence gives convergence of the ratios; if $\mathbb{P}(\mathsf{M}) = 0$, then every truncated hit probability is zero and all ratios are $+\infty$. Since capped experiments are a subset of all experiments, $\mathsf{SC}_b(Q) \le v_L(Q, b)$ for every L. Conversely, truncating any admissible experiment and passing to the limit gives the reverse inequality. Therefore

$$\mathsf{SC}_b(Q) = \inf_{L\ge 1} v_L(Q, b).$$

This is universally measurable by Lemma G.4. On $\{\mathsf{SC}_b < +\infty\}$, choose measurably the first L for which $v_L < \mathsf{SC}_b + \epsilon/2$, then use the capped $\epsilon/2$-selector. On $\{\mathsf{SC}_b = +\infty\}$, use any measurable one-query rule; the approximation statement is vacuous there. Countable patching proves the claim. $\square$

Proof of (40). Lemma G.5 selects, simultaneously at every active state, an experiment within ϵ of its pointwise infimum. Taking the worst state and then letting $\epsilon \downarrow 0$ gives $\mathsf{SC}^{\mathrm{unif}} \le \mathsf{SC}^{\mathrm{pt}}$. The reverse inequality is immediate from the definitions. For each fixed prior and finite global horizon, Lemma G.1 replaces the universal selector by a Borel policy with the same strategic law. $\square$

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References

Mridul Agarwal, Vaneet Aggarwal, Abhishek Kumar Umrawal, and Christopher J. Quinn. DART: Adaptive accept reject algorithm for non-linear combinatorial bandits. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 6557–6565, 2021. doi: 10.1609/aaai.v35i8.16812. URL https://ojs.aaai.org/index.php/AAAI/article/view/16812.

Kenneth J. Arrow, David Blackwell, and Meyer A. Girshick. Bayes and minimax solutions of sequential decision problems. *Econometrica*, 17(3/4):213–244, 1949. doi: 10.2307/1905525. URL https://doi.org/10.2307/1905525.

Dilip Arumugam and Benjamin Van Roy. Deciding what to learn: A rate-distortion approach. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 373–382. PMLR, 2021. URL https://proceedings.mlr.press/v139/arumugam21a.html.

Jean-Yves Audibert, Sébastien Bubeck, and Gábor Lugosi. Regret in online combinatorial optimization. *Mathematics of Operations Research*, 39(1):31–45, 2014. doi: 10.1287/moor.2013.0598. URL https://doi.org/10.1287/moor.2013.0598.

Alireza Bakhtiari, Tor Lattimore, and Csaba Szepesvári. Thompson sampling for bandit convex optimisation. In *Proceedings of the 38th Conference on Learning Theory*, volume 291 of *Proceedings of Machine Learning Research*, pages 231–263. PMLR, 2025. URL https://proceedings.mlr.press/v291/bakhtiari25a.html.

Dimitri P. Bertsekas and Steven E. Shreve. *Stochastic Optimal Control: The Discrete-Time Case*. Academic Press, New York, 1978.

Dimitri P. Bertsekas and John N. Tsitsiklis. An analysis of stochastic shortest path problems. *Mathematics of Operations Research*, 16(3):580–595, 1991. doi: 10.1287/moor.16.3.580. URL https://web.mit.edu/dimitrib/www/Stochasticsp.pdf.

David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032. URL https://doi.org/10.1214/aoms/1177729032.

Sébastien Bubeck and Mark Sellke. First-order Bayesian regret analysis of Thompson sampling. *IEEE Transactions on Information Theory*, 69(3):1795–1823, 2023. URL https://ieeexplore.ieee.org/document/9915622/.

Fan Chen, Dylan J. Foster, Yanjun Han, Jian Qian, Alexander Rakhlin, and Yunbei Xu. Assouad, Fano, and Le Cam with interaction: A unifying lower bound framework and characterization for bandit learnability. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2407. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/8a23a95e26d016711c0d70f79ade3c95-Abstract-Conference.html.

Herman Chernoff. Sequential design of experiments. *The Annals of Mathematical Statistics*, 30:755–770, 1959. doi: 10.1214/aoms/1177706205. URL https://doi.org/10.1214/aoms/1177706205.

Alon Cohen, Tamir Hazan, and Tomer Koren. Tight bounds for bandit combinatorial optimization. In *Proceedings of the 2017 Conference on Learning Theory*, volume 65 of *Proceedings of Machine Learning Research*, pages 629–642. PMLR, 2017. URL https://proceedings.mlr.press/v65/cohen17a.html.

Morris H. DeGroot. Uncertainty, information, and sequential experiments. *The Annals of Mathematical Statistics*, 33(2):404–419, 1962. doi: 10.1214/aoms/1177704567. URL https://doi.org/10.1214/aoms/1177704567.

Shi Dong and Benjamin Van Roy. An information-theoretic analysis for Thompson sampling with many actions. In *Advances in Neural Information Processing Systems 31*, pages 4161–4169, 2018. URL https://arxiv.org/abs/1805.11845.

Hidde Fokkema, Dirk van der Hoeven, Tor Lattimore, and Jack J. Mayo. Online Newton method for bandit convex optimisation. In *Proceedings of Thirty Seventh Conference on Learning Theory*, volume 247 of *Proceedings of Machine Learning Research*, pages 1713–1714. PMLR, 2024. URL https://proceedings.mlr.press/v247/fokkema24a.html. Full version: arXiv:2406.06506.

Dylan J. Foster, Sham M. Kakade, Jian Qian, and Alexander Rakhlin. The statistical complexity of interactive decision making, 2021. URL https://arxiv.org/abs/2112.13487.

Dylan J. Foster, Alexander Rakhlin, Ayush Sekhari, and Karthik Sridharan. On the complexity of adversarial decision making. In *Advances in Neural Information Processing Systems*, volume 35, pages 35404–35417, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/e5daf532498ebba4fe6544d49c811678-Abstract-Conference.html.

Dylan J. Foster, Noah Golowich, and Yanjun Han. Tight guarantees for interactive decision making with the decision-estimation coefficient. In *Proceedings of the 36th Conference on Learning Theory*, volume 195 of *Proceedings of Machine Learning Research*, pages 3969–4043. PMLR, 2023. URL https://proceedings.mlr.press/v195/foster23b.html.

Amaury Gouverneur, Borja Rodríguez-Gálvez, Tobias J. Oechtering, and Mikael Skoglund. Chained information-theoretic bounds and tight regret rate for linear bandit problems, 2024. URL https://arxiv.org/abs/2403.03361.

Shinji Ito, Daisuke Hatano, Hanna Sumita, Kei Takemura, Takuro Fukunaga, Naonori Kakimura, and Ken ichi Kawarabayashi. Improved regret bounds for bandit combinatorial optimization. In *Advances in Neural Information Processing Systems 32*, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/d5b3d8dadd770c460b1cde910a711987-Abstract.html.

Tor Lattimore and András György. Mirror descent and the information ratio. In *Proceedings of the 34th Conference on Learning Theory*, volume 134 of *Proceedings of Machine Learning Research*, pages 2965–2992. PMLR, 2021. URL https://proceedings.mlr.press/v134/lattimore21b.html.

Tor Lattimore and András György. A second-order method for stochastic bandit convex optimisation. In *Proceedings of the 36th Conference on Learning Theory*, volume 195 of *Proceedings of Machine Learning Research*, pages 2067–2094. PMLR, 2023. URL https://proceedings.mlr.press/v195/lattimore23a.html.

Dennis V. Lindley. On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics*, 27(4):986–1005, 1956. doi: 10.1214/aoms/1177728069. URL https://doi.org/10.1214/aoms/1177728069.

Arnab Maiti, Yunbei Xu, and Kevin Jamieson. On the power of adaptivity for ε-best arm identification in linear bandits. In *Proceedings of the Thirty-Ninth Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pages 4911–4968. PMLR, 2026. URL https://proceedings.mlr.press/v336/maiti26a.html.

Mohammad Naghshvar and Tara Javidi. Active sequential hypothesis testing. *The Annals of Statistics*, 41(6):2703–2738, 2013. doi: 10.1214/13-AOS1144. URL https://doi.org/10.1214/13-AOS1144.

Gergely Neu, Matteo Papini, and Ludovic Schwartz. Optimistic information directed sampling. In *Proceedings of the 37th Conference on Learning Theory*, volume 247 of *Proceedings of Machine Learning Research*, pages 3970–4006. PMLR, 2024. URL https://proceedings.mlr.press/v247/neu24a.html.

Nived Rajaraman. The price of hidden curvature: An $\widetilde{\Omega}(d^{5/4}\sqrt{T})$ lower bound for bandit convex optimization, 2026. URL https://arxiv.org/abs/2607.18652.

Idan Rejwan and Yishay Mansour. Top-k combinatorial bandits with full-bandit feedback. In *Proceedings of the 31st International Conference on Algorithmic Learning Theory*, volume 117 of *Proceedings of Machine Learning Research*, pages 752–776. PMLR, 2020. URL https://proceedings.mlr.press/v117/rejwan20a.html.

Daniel Russo and Benjamin Van Roy. An information-theoretic analysis of Thompson sampling. *Journal of Machine Learning Research*, 17(68):1–30, 2016. URL https://jmlr.org/papers/v17/14-087.html.

Daniel Russo and Benjamin Van Roy. Learning to optimize via information-directed sampling. *Operations Research*, 66(1):230–252, 2018. doi: 10.1287/opre.2017.1663. URL https://pubsonline.informs.org/doi/10.1287/opre.2017.1663.

Daniel Russo and Benjamin Van Roy. Satisficing in time-sensitive bandit learning. *Mathematics of Operations Research*, 47(4):2815–2839, 2022. doi: 10.1287/moor.2021.1229. URL https://doi.org/10.1287/moor.2021.1229.

Aadirupa Saha, Aldo Pacchiano, and Jonathan Lee. Dueling RL: Reinforcement learning with trajectory preferences. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pages 6263–6289. PMLR, 2023. URL https://proceedings.mlr.press/v206/saha23a.html.

Rajat Sen, Alexander Rakhlin, Lexing Ying, Rahul Kidambi, Dean Foster, Daniel N. Hill, and Inderjit S. Dhillon. Top-k extreme contextual bandits with arm hierarchy. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 9422–9433. PMLR, 2021. URL https://proceedings.mlr.press/v139/sen21a.html.

Maurice Sion. On general minimax theorems. *Pacific Journal of Mathematics*, 8(1):171–176, 1958. doi: 10.2140/pjm.1958.8.171. URL https://doi.org/10.2140/pjm.1958.8.171.

Yuanhao Wang, Qinghua Liu, and Chi Jin. Is RLHF more difficult than standard RL? a theoretical perspective. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/efb9629755e598c4f261c44aeb6fde5e-Abstract-Conference.html.

Yichong Xu, Ruosong Wang, Lin Yang, Aarti Singh, and Artur Dubrawski. Preference-based reinforcement learning with finite-time guarantees. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/d9d3837ee7981e8c064774da6cdd98bf-Abstract.html.

Yunbei Xu. Bellman-sufficient information complexity, 2026a. URL https://arxiv.org/abs/2606.11171.

Yunbei Xu. Pointwise complexity for Gaussian fields: Upper envelopes, algorithmic lower bounds, and separation, 2026b. URL https://arxiv.org/abs/2606.07931.

Yunbei Xu and Assaf Zeevi. Bayesian design principles for frequentist sequential learning. *Journal of the ACM*, 72(5):37:1–37:65, October 2025. doi: 10.1145/3766898. URL https://dl.acm.org/doi/10.1145/3766898.